Vector-potential finite-element formulations for two-dimensional resistive magnetohydrodynamics*

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Abstract

Vector-potential formulations are attractive for electromagnetic problems in two dimensions, since they reduce both the number and complexity of equations, particularly in coupled systems, such as magnetohydrodynamics (MHD). In this paper, we consider the finite-element formulation of a vector-potential model of two-dimensional resistive MHD. Existence and uniqueness are considered separately for the continuum nonlinear equations and the discretized and linearized form that arises from Newton's method applied to a modified system. Under some conditions, we prove that the solutions of the original and modified weak forms are the same, allowing us to prove convergence of both the discretization and the nonlinear iteration.

1 Introduction

Magnetohydrodynamics (MHD) models the flow of a charged fluid, or plasma, in the presence of electromagnetic fields. There are many formulations of MHD, depending on the domain and physical parameters considered. This includes assumptions associated with the coupling between the electric field, current density, and Ohm's law, leading to formulations such as ideal, resistive, and Hall MHD [13]. In this paper, we use a single incompressible fluid model, treating ions and electrons together, along with a resistive formulation. The resulting visco-resistive model couples the Navier-Stokes equations with Maxwell's equations, forming a nonlinear system of partial differential equations (PDEs). Moreover, we focus on time-independent solutions, with our primary focus on

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existence and uniqueness of solutions to the nonlinear and linearized systems of equations.

The equations of stationary, incompressible single fluid MHD posed in three dimensions are considered in (for example) [14, 15]. Under some conditions on the data, the existence and uniqueness of solutions to weak formulations of the equations is known both in the continuum and for certain discretizations. The focus of this paper is on MHD in two dimensions (2D). Here, a vector potential formulation, was used in [2, 8]. However, [2, 8] focus primarily on linear algebraic aspects of the solution of the resulting linearized systems of equations, leaving open the question of existence and uniqueness of solutions. In this paper, we focus on the theoretical analysis of the discretization. As such, we apply standard theory for the existence and uniqueness of solutions at both the continuum and discrete levels. Under some conditions, we prove that Newton's method yields well-posed linearizations and converges to the solution of the weak formulation. Notable in the analysis is the use of a modified weak form for the discretization, motivated by the existing work in [2, 8]. Furthermore, under some conditions, we prove that the solutions to the original and modified weak forms are the same. This validates the use of the modified formulation in practice.

An outline of this paper is as follows. In Section 2, we propose the vector-potential formulation for the MHD problem in 2D and, under certain conditions, we prove the existence and uniqueness of the continuum solution. In Section 3, we introduce a modified, "uncurled", formulation for the MHD problem and present the analysis of the discretized problem using a mixed finite-element method. In Section 4, we consider Newton's method for solving the nonlinear system and analyze convergence. Finally, some concluding remarks are given in Section 5.

In what follows, the letter C (with or without subscripts) denotes a generic positive constant which may be different depending on the context. For a Lipschitz domain $\Omega \subset \mathbb{R}^2$, denote by L^p, $1 \leq p \leq \infty$, the Lebesgue space of p-integrable functions, endowed with the norm $\|\cdot\|_{0,p}$. Denote the standard Euclidean norm as $|\cdot|$, the classical $L^2(\Omega)$ inner product and norm as $\langle\cdot,\cdot\rangle_0$ and $\|\cdot\|_0$, respectively, and $\langle f, g\rangle = \int_\Omega fg\mathrm{dX}$, where $fg \in L^1(\Omega)$. The standard L^2-based Sobolev space with integer or fractional exponent s is denoted by $H^s(\Omega)$. We write $\|\cdot\|_s$ for its norm.

For convenience, we introduce the spaces

$$\mathbf{J} := \left(H_0^1(\Omega)\right)^2 \cap H(\mathrm{div}^0;\Omega), \quad \mathbf{X} := H_\tau^2(\Omega), \quad \widetilde{\mathbf{X}} := H_\tau^1(\Omega)$$

$$\mathbf{W} := \left(H_0^1(\Omega)\right)^2, \quad \mathbf{Q} := L_0^2(\Omega),$$

where $\mathbf{J}$ and $\mathbf{X}$ are endowed with the norms, $\|\vec{v}\|_1, \|\phi\|_2$, respectively. Here,

$$\begin{aligned}
\left(H_0^1(\Omega)\right)^2 &:= \{\vec{v}\,|\,\vec{v} \in \left(H^1(\Omega)\right)^2, \vec{v}|_{\partial\Omega} = \vec{0}\}, \\
H(\mathrm{div}^0;\Omega) &:= \{\vec{v}\,|\,\vec{v} \in \left(L^2(\Omega)\right)^2, \nabla\cdot\vec{v} = 0 \text{ in } \Omega\},
\end{aligned}$$

$$
\begin{aligned}
H^2_\tau(\Omega) &:= \{\phi \,|\, \phi \in H^2(\Omega), \frac{\partial \phi}{\partial \vec{n}}|_{\partial\Omega} = 0, \int_\Omega \phi \,\mathrm{dX} = 0\}, \\
H^1_\tau(\Omega) &:= \{\phi \,|\, \phi \in H^1(\Omega), \frac{\partial \phi}{\partial \vec{n}}|_{\partial\Omega} = 0, \int_\Omega \phi \,\mathrm{dX} = 0\}, \\
L^2_0(\Omega) &:= \{q \,|\, q \in L^2(\Omega), \int_\Omega q \,\mathrm{dX} = 0\}.
\end{aligned}
$$

2 Steady-state visco-resistive MHD

In this paper, we consider cylindrical three-dimensional domains, $\hat{\Omega} = \Omega \times [z_0.z_1]$, where $\Omega \subset \mathbb{R}^2$ is Lipschitz, which are coupled with a large incident magnetic field in the z-direction. To begin, we consider the one-fluid visco-resistive MHD model, where the dependent variables are the fluid velocity $\vec{u}$, the hydrodynamic pressure p, and the magnetic field $\vec{B}$. The equations are

$$
\begin{aligned}
\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla)\vec{u} - \nabla \cdot (T + T_M) + \nabla p &= \vec{F}, && (1) \\
\frac{\partial \vec{B}}{\partial t} - \nabla \times (\vec{u} \times \vec{B}) + \nabla \times (\frac{1}{Re_m} \nabla \times \vec{B}) &= \vec{G}, && (2) \\
\nabla \cdot \vec{u} &= 0, && (3) \\
\nabla \cdot \vec{B} &= 0, && (4)
\end{aligned}
$$

where $\vec{G} = -\nabla \times \vec{E}_{\text{stat}}$, and $\vec{E}_{\text{stat}}$ is the static component of the electric field. The Newtonian and magnetic stress tensors are

$$
T = \frac{1}{2Re}\left[\nabla \vec{u} + \nabla \vec{u}^T\right], \text{ and } T_M = \vec{B} \otimes \vec{B} - \frac{1}{2}|\vec{B}|^2 I,
$$

respectively. We define the tensor $\vec{B} \otimes \vec{B}$ component-wise as $(\vec{B} \otimes \vec{B})_{i,j} = B_i B_j$ and $\vec{F} = (\vec{f}, 0) \in \left(H^{-1}(\hat{\Omega})\right)^3$ for $\vec{f} \in \left(H^{-1}(\Omega)\right)^2$, $\vec{G} \in \left(L^2(\hat{\Omega})\right)^3$. Additionally, we define the standard nondimensional Reynolds number, Re, and magnetic Reynolds number, Re_m:

$$
Re = \frac{\rho U L}{\nu}, \qquad Re_m = \frac{\mu_0 U L}{\eta},
$$

for a characteristic velocity, U, and a characteristic length scale, L. The physical parameters, all assumed constant, are the fluid viscosity ν, the fluid density ρ, the magnetic permeability of free space μ_0, and the magnetic resistivity η.

Assuming that the domain is coupled with a large incident magnetic field in the z-direction, the resulting dynamics decouple into a two-dimensional problem over Ω with simple behaviour in the z-direction. Thus, we consider the numerical solution of the resulting two-dimensional problem, which simulates a large aspect-ratio tokamak. Here, the magnetic field along the toroidal direction is very large in order to contain the plasma and, thus, we only consider plasma behavior in a poloidal cross-section. Assuming no variation in the toroidal (z)

direction, we take $\vec{B} = (B_1(x,y), B_2(x,y), B_0)$ and $\vec{u} = (u_1(x,y), u_2(x,y), u_0)$. Then, we complete the above system with the homogeneous boundary conditions

$$\vec{u} = \vec{0} \quad \text{on} \quad \partial\Omega, \tag{5}$$

$$\vec{n} \times \vec{B} = \vec{0} \quad \text{on} \quad \partial\Omega, \tag{6}$$

where $\vec{n}$ denotes the outward normal vector on $\partial\Omega$.

Noting that $\nabla \cdot \vec{B} = 0$, we must have $\frac{\partial B_1}{\partial x} + \frac{\partial B_2}{\partial y} = 0$, which allows us to write $\vec{B} = \nabla \times \vec{A} + (0, 0, B_0)$, where $\vec{A} = (0, 0, A(x,y))$. Consequently, we rewrite Equations (1)-(4) in terms of the vector potential, $\vec{A}$. Considering the continuum problem (1)-(4), direct calculation shows that B_0 and u_0 do not appear in the resulting equations for the other components of $\vec{B}$ and $\vec{u}$ and, so, we ignore them (by treating them as zero) in what follows.

2.1 The weak formulation

We now introduce the weak formulation of (1)-(4) for the two-dimensional domain Ω. If we write $\vec{B} = \nabla \times \vec{A}$ for a vector potential, $\vec{A}$, then $\nabla \cdot \vec{B} = 0$; as such, Equation (4) is automatically satisfied, and we no longer include it in the formulation.

A standard vector calculus identity is that if $\vec{B} \in \left(H^1(\hat{\Omega})\right)^3$,

$$\nabla \cdot (\vec{B} \otimes \vec{B} - \frac{1}{2}|\vec{B}|^2 I) = (\nabla \times \vec{B}) \times \vec{B} + (\nabla \cdot \vec{B}) \cdot \vec{B},$$

and if $\vec{B} \in \left(H^1(\hat{\Omega})\right)^3 \cap H(\mathrm{div}^0; \hat{\Omega})$, then

$$\nabla \cdot (\vec{B} \otimes \vec{B} - \frac{1}{2}|\vec{B}|^2 I) = (\nabla \times \vec{B}) \times \vec{B}.$$

Taking $\vec{B} = (\frac{\partial A}{\partial y}, -\frac{\partial A}{\partial x}, 0)$ ensures that $\vec{B} \in \left(H^1(\hat{\Omega})\right)^3 \cap H(\mathrm{div}^0; \hat{\Omega})$ when $A \in \mathbf{X}$, giving

$$\begin{aligned} \int_{\hat{\Omega}} \nabla \cdot (\vec{B} \otimes \vec{B} - \frac{1}{2}|\vec{B}|^2 I) \cdot \vec{V} \mathrm{d}\hat{\mathrm{X}} &= \int_{\hat{\Omega}} (\nabla \times \vec{B}) \times \vec{B} \cdot \vec{V} \mathrm{d}\hat{\mathrm{X}} \\ &= \int_{\hat{\Omega}} (-\triangle A \cdot \frac{\partial A}{\partial x}, -\triangle A \cdot \frac{\partial A}{\partial y}, 0) \cdot \vec{V} \mathrm{d}\hat{\mathrm{X}} \\ &= -(z_1 - z_0) \int_{\Omega} \triangle A \cdot (\nabla A \cdot \vec{v}) \, \mathrm{dX}, \end{aligned} \tag{7}$$

for any $\vec{V} = (\vec{v}, v_3) \in \left(H^1(\hat{\Omega})\right)^3$, with $\vec{v} \in \left(H^1(\Omega)\right)^2$.

Taking $\vec{C} = \nabla \times (0, 0, \varphi)$ for $\varphi \in \mathbf{X}$, then we can rewrite the weak formulation of (2), discarding the time derivative,

$$\int_{\hat{\Omega}} \left[-\nabla \times (\vec{u} \times \vec{B}) \cdot \vec{C} + \nabla \times (Re_m^{-1} \nabla \times \vec{B}) \cdot \vec{C} \right] \mathrm{d}\hat{\mathrm{X}} = \int_{\hat{\Omega}} \vec{G} \cdot \vec{C} \mathrm{d}\hat{\mathrm{X}},$$

as

$$\int_\Omega -(u_1,u_2)\cdot\nabla A\cdot\triangle\varphi\,\mathrm{dX}+\int_\Omega Re_m^{-1}\triangle A\cdot\triangle\varphi\,\mathrm{dX}=\int_\Omega E^0\cdot\triangle\varphi\mathrm{dX},$$

where E^0 is the z-component of the electrostatic part, $\vec{E}_{\text{stat}}$, and we choose E^0 so that $\int_\Omega E^0\mathrm{dX}=0$. We drop the common scaling of (z_1-z_0) when switching from integrals over $\hat{\Omega}$ to those over Ω. In the following, we denote $\vec{u}=(u_1(x,y),u_2(x,y))$.

Note that with $\vec{B}=\nabla\times(0,0,A(x,y))$, the natural boundary conditions enforced on the spaces $\mathbf{X}$ and $\widetilde{\mathbf{X}}$ imply tangential Dirichlet boundary conditions on B in (6). The weak form of (1)-(4) in two dimensions, coupled with boundary condition, (5)-(6), is : find $\vec{u}\in\mathbf{W},A\in\mathbf{X},p\in\mathbf{Q}$ such that

$$\begin{aligned} a_1(\vec{u},\vec{v})+c_0(\vec{u};\vec{u},\vec{v})+c_1(A;\vec{v},A)+b(p,\vec{v}) &= \langle\vec{f},\vec{v}\rangle, && (8)\\ a_2(A,\varphi)-c_1(A;\vec{u},\varphi) &= \langle E^0,\triangle\varphi\rangle, && (9)\\ b(q,\vec{u}) &= 0, && (10)\end{aligned}$$

for all $\vec{v}\in\mathbf{W},\varphi\in\mathbf{X},q\in\mathbf{Q}$, with $\mathcal{S}\vec{u}=\frac{1}{2}(\nabla\vec{u}+\nabla\vec{u}^T)$, where

$$\begin{aligned} a_1(\vec{u},\vec{v}) &:= Re^{-1}\int_\Omega\mathcal{S}\vec{u}:\nabla\vec{v}\,\mathrm{dX}=Re^{-1}\int_\Omega\mathcal{S}\vec{u}:\mathcal{S}\vec{v}\,\mathrm{dX},\\ a_2(\phi,\psi) &:= Re_m^{-1}\int_\Omega\triangle\phi\cdot\triangle\psi\,\mathrm{dX},\\ b(q,\vec{v}) &:= -\int_\Omega q(\nabla\cdot\vec{v})\,\mathrm{dX},\\ c_0(\vec{w};\vec{u},\vec{v}) &:= \int_\Omega(\vec{w}\cdot\nabla)\vec{u}\cdot\vec{v}\,\mathrm{dX},\\ c_1(\psi;\vec{v},\phi) &:= \int_\Omega\triangle\phi\cdot\nabla\psi\cdot\vec{v}\,\mathrm{dX}.\end{aligned}$$

2.2 Properties of the weak formulation

For later analysis, we state the standard compact imbedding theorem for Sobolev spaces.

Proposition 1 ([12], Theorem I.1.2)**.** *Let Ω be a bounded and open Lipschitz-continuous subset of $\mathbb{R}^N$ and let $p\in\mathbb{R}$ with $1\le p<\infty$, and $m,n\in\mathbb{N}$ with $n\le m$. The imbedding of $W^{m,p}(\Omega)$ into $W^{n,q}(\Omega)$ is compact for all real q that satisfy:*

$$1\le q<\infty,\quad \textit{when}\quad N=(m-n)p.$$

Next, we show the trilinear forms c_0 and c_1 are well defined.

Lemma 1. *If $\vec{u},\vec{v},\vec{w}\in\left(H^1(\Omega)\right)^2$, then*

$$|c_0(\vec{w};\vec{u},\vec{v})|\le C_0\|\vec{w}\|_{0,4}\cdot\|\nabla\vec{u}\|_0\cdot\|\vec{v}\|_{0,4}\le C_0\|\vec{w}\|_1\cdot\|\vec{u}\|_1\cdot\|\vec{v}\|_1, \tag{11}$$

where C_0 is a constant depending only on Ω.

Proof: According to Proposition 1, the space $H^1(\Omega)$ is continuously imbedded in $L^4(\Omega)$. Then, by Hölder's inequality, for $\vec{u}, \vec{v}, \vec{w} \in \left(H^1(\Omega)\right)^2$, we have

$$w_j \cdot (D_j u_i) \cdot v_i \in L^1(\Omega), \quad i, j = 1, 2,$$

where $D_j u_i$ is the derivative of u_i with respect to the j-th component, for $\vec{u} = (u_1, u_2)$, and

$$\begin{aligned} \left| \int_\Omega w_j \cdot (D_j u_i) \cdot v_i \, \mathrm{dX} \right| &\leq C_0 \|w_j\|_{0,4} \cdot \|D_j u_i\|_0 \cdot \|v_i\|_{0,4} \\ &\leq C_0 \|w_j\|_1 \cdot |u_i|_1 \cdot \|v_i\|_1. \end{aligned}$$

Since

$$|c_0(\vec{w}; \vec{u}, \vec{v})| = \left| \int_\Omega \sum_{i,j=1}^{2} w_j \cdot (D_j u_i) \cdot v_i \, \mathrm{dX} \right|,$$

we have

$$\begin{aligned} |c_0(\vec{w}; \vec{u}, \vec{v})| &\leq C_0 \|\vec{w}\|_1 |\vec{u}|_1 \|\vec{v}\|_1 \\ &\leq C_0 \|\vec{w}\|_1 \|\vec{u}\|_1 \|\vec{v}\|_1. \end{aligned}$$

□

Lemma 2. *If $\psi, \phi \in H^2(\Omega)$ and $\vec{v} \in \left(H^1(\Omega)\right)^2$, then*

$$|c_1(\psi; \vec{v}, \phi)| \leq C_1 \|\nabla \psi\|_{0,4} \cdot \|\triangle \phi\|_0 \cdot \|\vec{v}\|_{0,4} \leq C_1 \|\psi\|_2 \cdot \|\phi\|_2 \cdot \|\vec{v}\|_1, \tag{12}$$

where C_1 is a constant depending only on Ω.

Proof: Let $\vec{v} = (v_1, v_2)$, then

$$v_i \cdot (D_i \psi) \cdot \triangle \phi \in L^1(\Omega), \quad i = 1, 2.$$

By Hölder's inequality and the property that $v_i, D_i \psi \in H^1(\Omega)$ are compactly imbedded into $L^4(\Omega)$, we have

$$\begin{aligned} \left| \int_\Omega v_i \cdot (D_i \psi) \cdot \triangle \phi \, \mathrm{dX} \right| &\leq \|v_i\|_{0,4} \cdot \|(D_i \psi)\|_{0,4} \cdot \|\triangle \phi\|_0 \\ &\leq C_1 \|v_i\|_{0,4} \cdot \|\nabla \psi\|_{0,4} \cdot \|\triangle \phi\|_0 \\ &\leq C_1 \|v_i\|_{0,4} \cdot \|\nabla \psi\|_1 \cdot \|\triangle \phi\|_0. \end{aligned}$$

Furthermore,

$$\begin{aligned} |c_1(\psi; \vec{v}, \phi)| &= \left| \int_\Omega \vec{v} \cdot \nabla \psi \cdot \triangle \phi \, \mathrm{dX} \right| \\ &= \left| \int_\Omega \left[v_1 \cdot (D_1 \psi) \cdot \triangle \phi + v_2 \cdot (D_2 \psi) \cdot \triangle \phi \right] \mathrm{dX} \right| \end{aligned}$$

$$
\begin{aligned}
&\leq C_1\big(\|v_1\|_{0,4}+\|v_2\|_{0,4}\big)\cdot\|\nabla\psi\|_1\cdot\|\triangle\phi\|_0\\
&\leq C_1\|\vec{v}\|_{0,4}\cdot\|\nabla\psi\|_1\cdot\|\triangle\phi\|_0\\
&\leq C_1\|\vec{v}\|_1\cdot\|\psi\|_2\cdot\|\phi\|_2.
\end{aligned}
$$

□

Lemma 3 ([16], Lemma II.1.3)**.** *Let $\vec{u}\in\mathbf{J}$, and $\vec{w},\vec{v}\in\big(H^1(\Omega)\big)^2$, then the following results hold:*

$$
\begin{aligned}
c_0(\vec{u};\vec{v},\vec{v}) &= 0,\\
c_0(\vec{u};\vec{w},\vec{v}) &= -c_0(\vec{u};\vec{v},\vec{w}).
\end{aligned}
\tag{13}
$$

Proof: See [16]. □ Putting aside the pressure p for a moment, we define an auxiliary variational formulation of (1)-(4) based on the spaces $\mathbf{J}$ and $\mathbf{X}$.

Formulation 1. *Find $(\vec{u},A)\in\mathbf{J}\times\mathbf{X}$ such that*

$$
\begin{aligned}
a_1(\vec{u},\vec{v})+c_0(\vec{u};\vec{u},\vec{v})+c_1(A;\vec{v},A) &= \langle\vec{f},\vec{v}\rangle,\\
a_2(A,\varphi)-c_1(A;\vec{u},\varphi) &= \langle E^0,\triangle\varphi\rangle,
\end{aligned}
$$

for all $(\vec{v},\varphi)\in\mathbf{J}\times\mathbf{X}$.

For our analysis, we write Formulation 1 in compact form: find $(\vec{u},A)\in\mathbf{J}\times\mathbf{X}$, such that

$$
\mathcal{A}(\vec{u},A;\vec{v},\varphi)+\mathcal{C}(\vec{u},A;\vec{u},A;\vec{v},\varphi)=\mathcal{L}(\vec{v},\varphi),
$$

for all $(\vec{v},\varphi)\in\mathbf{J}\times\mathbf{X}$. Here,

$$
\begin{aligned}
\mathcal{A}(\vec{u},A;\vec{v},\varphi) &:= a_1(\vec{u},\vec{v})+a_2(A,\varphi),\\
\mathcal{C}(\vec{w},\psi;\vec{u},\phi;\vec{v},\varphi) &:= c_0(\vec{w};\vec{u},\vec{v})+c_1(\psi;\vec{v},\phi)-c_1(\psi;\vec{u},\varphi),\\
\mathcal{L}(\vec{v},\varphi) &:= \langle\vec{f},\vec{v}\rangle+\langle E^0,\triangle\varphi\rangle.
\end{aligned}
$$

We define the product space $\mathbf{W}\times\mathbf{X}$ with the norm $|||(\vec{v},\varphi)|||^2:=\|\vec{v}\|_1^2+\|\varphi\|_2^2$ and define the operator norm, $|||\mathcal{L}|||_-:=\sup\limits_{(\vec{0},0)\neq(\vec{v},\varphi)\in\mathbf{J}\times\mathbf{X}}\dfrac{\mathcal{L}(\vec{v},\varphi)}{|||(\vec{v},\varphi)|||}$.

Next, we consider the properties of the forms $\mathcal{A},\mathcal{C}$.

Lemma 4. *For any $(\vec{v},\varphi),(\vec{w},\psi)\in\mathbf{W}\times\mathbf{X}$, we have*

$$
\begin{aligned}
\mathcal{A}(\vec{v},\varphi;\vec{v},\varphi) &\geq c_\alpha\min\{Re^{-1},Re_m^{-1}\}|||(\vec{v},\varphi)|||^2,\\
\mathcal{A}(\vec{w},\psi;\vec{v},\varphi) &\leq \max\{2Re^{-1},Re_m^{-1}\}|||(\vec{w},\psi)|||\cdot|||(\vec{v},\varphi)|||,
\end{aligned}
\tag{14}
$$

where $c_\alpha\leq 1$ is a constant depending only on Ω.

Proof: Since $(\vec{v},\varphi)\in\mathbf{W}\times\mathbf{X}$, we have

$$
\mathcal{A}(\vec{v},\varphi;\vec{v},\varphi) = Re^{-1}\int_\Omega\mathcal{S}\vec{v}:\mathcal{S}\vec{v}\,\mathrm{dX}+\int_\Omega Re_m^{-1}\triangle\varphi\cdot\triangle\varphi\,\mathrm{dX}
$$

$$
\begin{aligned}
&= Re^{-1}\|\mathcal{S}\vec{v}\|_0^2 + Re_m^{-1}\|\triangle\varphi\|_0^2 \\
&\geq \beta_1 Re^{-1}\|\vec{v}\|_1^2 + \beta_2 Re_m^{-1}\|\varphi\|_2^2 \\
&\geq c_\alpha \min\{Re^{-1}, Re_m^{-1}\}|||(\vec{v},\varphi)|||^2,
\end{aligned}
$$

where $c_\alpha = \min\{\beta_1, \beta_2\}$, β_1 comes from Korn's Inequality, cf.[5, Corollary 11.2.22], and β_2 comes from a regularity argument, cf.[12, Section I, Theorem 1.10]. This gives the coercivity of $\mathcal{A}$.

For continuity,

$$
\begin{aligned}
\mathcal{A}(\vec{u},\psi;\vec{v},\varphi) &= Re^{-1}\int_\Omega \mathcal{S}\vec{u} : \mathcal{S}\vec{v}\,\mathrm{dX} + Re_m^{-1}\int_\Omega \triangle\psi\cdot\triangle\varphi\,\mathrm{dX} \\
&\leq 2Re^{-1}\|\vec{u}\|_1\|\vec{v}\|_1 + Re_m^{-1}\|\psi\|_2\|\varphi\|_2 \\
&\leq \max\{2Re^{-1}, Re_m^{-1}\}|||(\vec{u},\psi)|||\cdot|||(\vec{v},\varphi)|||,
\end{aligned}
$$

via the Cauchy-Schwarz inequality. □

Lemma 5. *For any $\vec{w},\vec{u},\vec{v}\in\mathbf{W}$ and $\psi,\phi,\varphi\in\mathbf{X}$, the trilinear form $\mathcal{C}$ has the following properties*

$$
|\mathcal{C}(\vec{w},\psi;\vec{u},\phi;\vec{v},\varphi)| \leq C_c|||(\vec{w},\psi)|||\cdot|||(\vec{u},\phi)|||\cdot|||(\vec{v},\varphi)|||, \tag{15}
$$

where C_c is a constant only depending on Ω.

Furthermore, if $\vec{w}\in\mathbf{J}$, we have

$$
\mathcal{C}(\vec{w},\psi;\vec{v},\varphi;\vec{v},\varphi) = 0. \tag{16}
$$

Proof: The first inequality is a result of inequalities (11) and (12). The second equality is due to Equality (13) and the definition of c_1. □

2.3 Mixed variational formulation

In this section, we will analyze the mixed variational formulation (8)-(10). First, recall the form $b:\mathbf{Q}\times\mathbf{W}\to\mathbb{R}$ given by:

$$
b(q,\vec{v}) = -\int_\Omega q(\nabla\cdot\vec{v})\,\mathrm{dX}.
$$

This form is continuous and satisfies the following inf-sup condition

$$
\inf_{0\neq q\in\mathbf{Q}}\sup_{\vec{0}\neq\vec{v}\in\mathbf{W}}\frac{b(q,\vec{v})}{\|\vec{v}\|_1\|q\|_0} \geq \Gamma > 0, \tag{17}
$$

where Γ is a constant depending only on Ω . The inf-sup condition for the form b is found, for example, in [12, Chapter I.5.1].

We introduce the extended form

$$
\mathcal{B}(q;\vec{v},\varphi) := b(q,\vec{v}),
$$

and define the kernel

$$\mathbf{Z} := \{\vec{v} \in \mathbf{W} : b(q, \vec{v}) = 0, \forall q \in \mathbf{Q}\}.$$

In the continuum, it is obvious that $\mathbf{Z} = \mathbf{J}$, given appropriate boundary conditions. The form $\mathcal{B}$ is also continuous:

$$|\mathcal{B}(q; \vec{v}, \varphi)| \leq C_b \|q\|_0 \|\vec{v}\|_1 \leq C_b \|q\|_0 |||(\vec{v}, \varphi)|||,$$

for all $(\vec{v}, q, \varphi) \in \mathbf{W} \times \mathbf{Q} \times \mathbf{X}$, with a constant $C_b > 0$. Furthermore, we have the inf-sup condition for $\mathcal{B}$.

Lemma 6. *There exists a constant $\Gamma > 0$ depending only on Ω, such that*

$$\sup_{(\vec{0},0) \neq (\vec{v},\varphi) \in \mathbf{W} \times \mathbf{X}} \frac{\mathcal{B}(q; \vec{v}, \varphi)}{|||(\vec{v}, \varphi)|||} \geq \Gamma \|q\|_0,$$

for all $q \in \mathbf{Q}$.

Proof: Since

$$\mathcal{B}(q; \vec{v}, \varphi) = b(q, \vec{v}),$$

we have

$$\sup_{(\vec{0},0) \neq (\vec{v},\varphi) \in \mathbf{W} \times \mathbf{X}} \frac{\mathcal{B}(q; \vec{v}, \varphi)}{|||(\vec{v}, \varphi)|||} \geq \sup_{\vec{0} \neq \vec{v} \in \mathbf{W}} \frac{b(q, \vec{v})}{\|\vec{v}\|_1} \geq \|q\|_0 \cdot \Gamma,$$

where the last inequality follows directly from (17). □ The mixed variational formulation (8)-(10) has the following formulation:

Formulation 2. *Find $(\vec{u}, p, A) \in \mathbf{W} \times \mathbf{Q} \times \mathbf{X}$ such that*

$$\mathcal{A}(\vec{u}, A; \vec{v}, \varphi) + \mathcal{C}(\vec{u}, A; \vec{u}, A; \vec{v}, \varphi) + \mathcal{B}(p; \vec{v}, \varphi) = \mathcal{L}(\vec{v}, \varphi), \tag{18}$$

$$\mathcal{B}(q; \vec{u}, A) = 0, \tag{19}$$

for all $(\vec{v}, q, \varphi) \in \mathbf{W} \times \mathbf{Q} \times \mathbf{X}$.

2.4 Existence and uniqueness of solutions

From [12], we quote the main theorem that we will apply to Formulations 1 and 2.

Theorem 1 ([12], Theorem IV.1.3)**.** *Let V be a separable Hilbert space with the norm $\|\cdot\|_V$, l be a linear functional in the dual space V' and, for $w \in V$, the mapping $(u, v) \to a(w; u, v)$ be a bilinear continuous form on $V \times V$. Assume that the following hold:*

- *the bilinear form $a(w; v, v)$ is uniformly V-coercive with respect to w, i.e., there exists a constant $\alpha > 0$ such that*

$$a(w; v, v) \geq \alpha \|v\|_V^2, \quad \forall v, w \in V;$$

- *there exists a continuous and monotonically increasing function* $L : \mathbb{R}_+ \to \mathbb{R}_+$ *such that for all* $\mu > 0$

$$|a(w_1; u, v) - a(w_2; u, v)| \leq L(\mu)\|u\|_V \|v\|_V \|w_1 - w_2\|_V,$$

$$\forall u, v \in V, \quad w_1, w_2 \in S_\mu = \{w \in V; \|w\|_V \leq \mu\}.$$

- *the linear function* l *and* α *satisfy*

$$\frac{\|l\|_{V'}}{\alpha^2} \cdot L(\|l\|_{V'}/\alpha) < 1.$$

Then the problem: find $u \in V$ *such that*

$$a(u; u, v) = l(v), \quad \forall v \in V,$$

has a unique solution that satisfies the stability bound $\|u\|_V \leq \alpha^{-1}\|l\|_{V'}$.

Theorem 2. *For* $\vec{f} \in \left(H^{-1}(\Omega)\right)^2$ *and* $E^0 \in L^2(\Omega)$*, assume that*

$$\frac{C_c |||\mathcal{L}|||_-}{c_\alpha^2 \min\{Re^{-2}, Re_m^{-2}\}} < 1, \tag{20}$$

where c_α *comes from (14), and* C_c *comes from (15).*

Then there exists a unique solution $(\vec{u}, A)$ *in* $\mathbf{J} \times \mathbf{X}$ *of the variational problem in Formulation 1. Furthermore, we have the stability bound*

$$|||(\vec{u}, A)||| \leq \frac{|||\mathcal{L}|||_-}{c_\alpha \min\{Re^{-1}, Re_m^{-1}\}}.$$

Proof: Step 1: We show that the space $\mathbf{J} \times \mathbf{X}$ is separable. Since $\mathbf{J}$ and $\mathbf{X}$ are closed subsets of $\left(H^1(\Omega)\right)^2$ and $H^2(\Omega)$ respectively, and $\left(H^1(\Omega)\right)^2$ and $H^2(\Omega)$ are separable Hilbert Spaces, $\mathbf{J}$ and $\mathbf{X}$ are separable.

Step 2: For any $(\vec{w}, \psi)$, define the mapping $((\vec{u}, \phi), (\vec{v}, \varphi)) \to \mathcal{A}_1(\vec{w}, \psi; \vec{u}, \phi, \vec{v}, \varphi)$, where

$$\mathcal{A}_1(\vec{w}, \psi; \vec{u}, \phi, \vec{v}, \varphi) = \mathcal{A}(\vec{u}, \phi; \vec{v}, \varphi) + \mathcal{C}(\vec{w}, \psi; \vec{u}, \phi; \vec{v}, \varphi).$$

Next, we show that

$$|\mathcal{A}_1(\vec{w}, \psi; \vec{v}, \varphi; \vec{v}, \varphi)| \geq \alpha |||(\vec{v}, \varphi)|||^2, \quad \forall (\vec{w}, \psi), (\vec{v}, \varphi) \in \mathbf{J} \times \mathbf{X}.$$

This results from inequalities (14) and (16),

$$\begin{aligned} |\mathcal{A}_1(\vec{w}, \psi; \vec{v}, \varphi; \vec{v}, \varphi)| &= |\mathcal{A}(\vec{v}, \varphi; \vec{v}, \varphi) + \mathcal{C}(\vec{w}, \psi; \vec{v}, \varphi; \vec{v}, \varphi)| \\ &= |\mathcal{A}(\vec{v}, \varphi; \vec{v}, \varphi)| \\ &\geq c_\alpha \min\{Re^{-1}, Re_m^{-1}\} |||(\vec{v}, \varphi)|||^2. \end{aligned}$$

Let $\alpha = c_\alpha \min\{Re^{-1}, Re_m^{-1}\}$.

Step 3: Linearity in the first argument of $\mathcal{C}$ and inequality (15) give

$$\begin{aligned}
|\mathcal{A}_1(\vec{w}_1, \psi_1; \vec{u}, \phi; \vec{v}, \varphi) &- \mathcal{A}_1(\vec{w}_2, \psi_2; \vec{u}, \phi; \vec{v}, \varphi)| \\
&= |\mathcal{C}((\vec{w}_1, \psi_1; \vec{u}, \phi; \vec{v}, \varphi) - \mathcal{C}(\vec{w}_2, \psi_2; \vec{u}, \phi; \vec{v}, \varphi)| \\
&= |\mathcal{C}(\vec{w}_1 - \vec{w}_2, \psi_1 - \psi_2; \vec{u}, \phi; \vec{v}, \varphi)| \\
&\leq C_c |||(\vec{w}_1 - \vec{w}_2, \psi_1 - \psi_2)||| \cdot |||(\vec{u}, \phi)||| \cdot |||(\vec{v}, \varphi)|||,
\end{aligned}$$

$\forall (\vec{w}_1, \psi_1), (\vec{w}_2, \psi_2), (\vec{u}, \phi), (\vec{v}, \varphi) \in \mathbf{J} \times \mathbf{X}$. Let $L(\mu) = C_c$, where C_c comes from (15).

Thus from Theorem 1, assumption (20) proves existence of a unique solution to Formulation 1. □

Finally, we consider the estimates for the unique solution of the MHD problem.

Theorem 3. *Let $\vec{f} \in \left(H^{-1}(\Omega)\right)^2$ and $E^0 \in L^2(\Omega)$, and assume that (20) holds. Then, there exists a unique solution $(\vec{u}, p, A)$ in $\mathbf{W} \times \mathbf{Q} \times \mathbf{X}$ of Formulation 2. Furthermore, we have the stability bound*

$$|||(\vec{u}, A)||| \leq \frac{|||\mathcal{L}|||_-}{c_\alpha \min\{Re^{-1}, Re_m^{-1}\}}$$

and

$$\|p\|_0 \leq \Gamma^{-1}\left[||\vec{f}||_{-1} + 2Re^{-1}\|\vec{u}\|_1 + C_0\|\vec{u}\|_1^2 + C_1\|A\|_1^2\right],$$

where C_0 comes from (11), and C_1 comes from (12).

Proof: Noting that $b(q, \vec{v}) = 0$ for all $q \in \mathbf{Q}$ when $\vec{v} \in \mathbf{J}$. From the properties of $\mathcal{B}$, restricting $\vec{v}$ in Formulation 2 to be in $\mathbf{J}$, we see that Formulation 2 reduces to Formulation 1, which we know has a unique solution. Let $(\vec{u}, A) \in \mathbf{J} \times \mathbf{X}$ be the unique solution of the variational problem in Formulation 1. According to Theorem 2, we have the bound on $(\vec{u}, A)$.

By the inf-sup condition in Lemma 17, there exists a unique solution of the following problem: find $p \in \mathbf{Q}$ such that

$$\begin{aligned}
b(p, \vec{v}) = \mathcal{B}(p; \vec{v}, \varphi) &= \mathcal{L}(\vec{v}, \varphi) - \mathcal{A}(\vec{u}, A; \vec{v}, \varphi) - \mathcal{C}(\vec{u}, A; \vec{u}, A; \vec{v}, \varphi), \\
&= \langle \vec{f}, \vec{v} \rangle - a_1(\vec{u}, \vec{v}) - c_0(\vec{u}; \vec{u}, \vec{v}) - c_1(A; \vec{v}, A),
\end{aligned}$$

for all $\vec{v} \in \mathbf{W} \setminus \mathbf{J}$. The details can be found in [12, Theorem IV.1.4].

From Equations (17), we have

$$\begin{aligned}
\Gamma\|p\|_0 &\leq \sup_{\vec{0} \neq \vec{v} \in \mathbf{W}} \frac{b(q, \vec{v})}{\|\vec{v}\|_1} \\
&= \sup_{\vec{0} \neq \vec{v} \in \mathbf{W}} \frac{\langle \vec{f}, \vec{v} \rangle - a_1(\vec{u}, \vec{v}) - c_0(\vec{u}; \vec{u}, \vec{v}) - c_1(A; \vec{v}, A)}{\|\vec{v}\|_1}.
\end{aligned}$$

Combining this with Equations (11) and (12), we obtain the bound on p. □

Any conforming mixed finite-element discretization of (18) and (19) necessarily

requires the use of H^2-conforming elements for $A \in \mathbf{X}$, such as Argyris triangle elements, or Bogner-Fox-Schmit elements [7]. While these approximations have been thoroughly studied, particularly for fourth-order problems, their use also poses some additional difficulties for implementation. Thus, we next consider a nonconforming approach using H^1-conforming elements, following [2, 8].

3 Uncurled formulation of MHD

To start, we consider a different formulation of Equation (2), noting that, in the steady-state case, it can be rewritten as $\nabla\times(-\vec{u}\times\vec{B}+Re_m^{-1}\nabla\times\vec{B}) = -\nabla\times\vec{E}_{\text{stat}}$. Thus, we can rewrite (2) as a first-order equation in $\vec{B}$, resulting in a second-order equation in A. Using this in place of (2), we derive a 2D weak formulation: find $(\vec{u}, A) \in \mathbf{W} \times \widetilde{\mathbf{X}}, p \in \mathbf{Q}$ such that

$$
\begin{aligned}
a_1(\vec{u},\vec{v}) + c_0(\vec{u};\vec{u},\vec{v}) + \widetilde{c}_1(A;\vec{v},A) + b(p,\vec{v}) &= \langle \vec{f},\vec{v}\rangle, && (21)\\
\widetilde{a}_2(A,\psi) + \widetilde{c}_2(A;\vec{u},\psi) &= \langle -E^0,\psi\rangle, && (22)\\
b(q,\vec{u}) &= 0, && (23)
\end{aligned}
$$

for all $(\vec{v},\psi) \in \mathbf{W} \times \widetilde{\mathbf{X}}, q \in \mathbf{Q}$, where

$$
\begin{aligned}
\widetilde{a}_2(\phi,\psi) &:= Re_m^{-1}\int_\Omega \nabla\phi\cdot\nabla\psi\,\mathrm{dX},\\
\widetilde{c}_1(\phi;\vec{v},A) &:= \frac{1}{2}\left\langle\left(\frac{\partial A}{\partial y}\cdot\frac{\partial\phi}{\partial y} - \frac{\partial A}{\partial x}\cdot\frac{\partial\phi}{\partial x}, -\left[\frac{\partial A}{\partial x}\cdot\frac{\partial\phi}{\partial y} + \frac{\partial A}{\partial x}\cdot\frac{\partial\phi}{\partial y}\right]\right), \frac{\partial\vec{v}}{\partial x}\right\rangle_0,\\
&\quad + \frac{1}{2}\left\langle\left(-\left[\frac{\partial A}{\partial x}\cdot\frac{\partial\phi}{\partial y} + \frac{\partial A}{\partial x}\cdot\frac{\partial\phi}{\partial y}\right], \frac{\partial A}{\partial x}\cdot\frac{\partial\phi}{\partial x} - \frac{\partial A}{\partial y}\cdot\frac{\partial\phi}{\partial y}\right), \frac{\partial\vec{v}}{\partial y}\right\rangle_0,\\
\widetilde{c}_2(\phi;\vec{u},\psi) &:= \int_\Omega \vec{u}\cdot\nabla\phi\cdot\psi\,\mathrm{dX}.
\end{aligned}
$$

Note, we now integrate by parts on the stress tensor in (1). This formulation is used in [2, 8]; in [2], an inf-sup stable FEM pair is used for discretization of $\vec{u}$ and p, while a stabilized pair was used in [8]. Neither of these papers considered theoretical analysis of this formulation, which we do here.

The analysis below shows that, in contrast to the formulation considered above, this formulation does not directly yield unique solutions under the classical theory. To address this, we augment analysis of the continuum weak form with that at the discrete level. We separately consider the well-posedness of the Newton linearizations in Section 4.

3.1 Properties of weak formulation

Recall the space defined previously,

$$
\widetilde{\mathbf{X}} = \{\phi\,|\phi \in H^1(\Omega), \frac{\partial\phi}{\partial\vec{n}}|_{\partial\Omega} = 0, \int_\Omega \phi\,\mathrm{dX} = 0\}.
$$

Given the difficulties in implementing an H^2-conforming discretization scheme, we no longer want to approximate $A \in H^2(\Omega)$. In the following, we consider our problem in $\widetilde{\mathbf{X}}$. Thus, we define an auxiliary variational formulation related to (21)-(22) based on the spaces $\mathbf{J}$ and $\widetilde{\mathbf{X}}$.

Formulation 3. *Find* $(\vec{u}, A) \in \mathbf{J} \times \widetilde{\mathbf{X}}$ *such that*

$$\begin{aligned} a_1(\vec{u},\vec{v}) + c_0(\vec{u};\vec{u},\vec{v}) + \widetilde{c}_1(A;\vec{v},A) &= \langle \vec{f},\vec{v}\rangle, \\ \widetilde{a}_2(A,\psi) + \widetilde{c}_2(A;\vec{u},\psi) &= \langle -E^0,\psi\rangle, \end{aligned}$$

for all $(\vec{v},\psi) \in \mathbf{J} \times \widetilde{\mathbf{X}}$.

For our analysis, we write Formulation 3 in compact form: find $(\vec{u}, A) \in \mathbf{J} \times \widetilde{\mathbf{X}}$, such that

$$\widetilde{\mathcal{A}}(\vec{u},A;\vec{v},\psi) + \widetilde{\mathcal{C}}(\vec{u},A;\vec{u},A;\vec{v},\psi) = \widetilde{\mathcal{L}}(\vec{v},\psi),$$

for all $(\vec{v},\psi) \in \mathbf{J} \times \widetilde{\mathbf{X}}$. Here,

$$\begin{aligned} \widetilde{\mathcal{A}}(\vec{u},A;\vec{v},\psi) &:= a_1(\vec{u},\vec{v}) + \widetilde{a}_2(A,\psi), \\ \widetilde{\mathcal{C}}(\vec{w},\phi;\vec{u},A;\vec{v},\psi) &:= c_0(\vec{w};\vec{u},\vec{v}) + \widetilde{c}_1(\psi;\vec{v},A) + \widetilde{c}_2(\psi;\vec{u},\phi), \\ \widetilde{\mathcal{L}}(\vec{v},\psi) &:= \langle \vec{f},\vec{v}\rangle + \langle -E^0,\psi\rangle. \end{aligned}$$

Lemma 7. *Let* $\psi, \phi \in H^1(\Omega)$ *and* $\vec{u} \in \left(H^1(\Omega)\right)^2$*, then*

$$|\widetilde{c}_2(\phi;\vec{u},\psi)| \le C\|\vec{u}\|_{0,4} \cdot \|\nabla\phi\|_0 \cdot \|\psi\|_{0,4} \le C\|\vec{u}\|_1 \cdot \|\phi\|_1 \cdot \|\psi\|_1,$$

where C *is a constant depending only on* Ω.

Proof: The proof is analogous to that in Lemma 2. □

We define the product space $\mathbf{W} \times \widetilde{\mathbf{X}}$ with the norm

$$\|(\vec{v},\psi)\|_1^2 := \|\vec{v}\|_1^2 + \|\psi\|_1^2,$$

and consider ellipticity of $\widetilde{\mathcal{A}}$ on this product space.

Lemma 8. *For any* $(\vec{v},\varphi) \in \mathbf{W} \times \widetilde{\mathbf{X}}$*, we have*

$$\begin{aligned} \widetilde{\mathcal{A}}(\vec{v},\varphi;\vec{v},\varphi) &\ge \widetilde{c}_\alpha \min\{Re^{-1}, Re_m^{-1}\}\|(\vec{v},\varphi)\|_1^2, \\ \widetilde{\mathcal{A}}(\vec{w},\psi;\vec{v},\varphi) &\le \max\{2Re^{-1}, Re_m^{-1}\}\|(\vec{w},\psi)\|_1\|(\vec{v},\varphi)\|_1, \end{aligned}$$

where $\widetilde{c}_\alpha \le 1$ *is a constant depending only on* Ω.

Proof: For any $(\vec{v},\varphi) \in \mathbf{W} \times \widetilde{\mathbf{X}}$, we have

$$\begin{aligned} \widetilde{\mathcal{A}}(\vec{v},\varphi;\vec{v},\varphi) &= Re^{-1}\int_\Omega \mathcal{S}\vec{v} : \mathcal{S}\vec{v}\,\mathrm{dX} + \int_\Omega Re_m^{-1}\nabla\varphi\cdot\nabla\varphi\,\mathrm{dX} \\ &= Re^{-1}\|\mathcal{S}\vec{v}\|_0^2 + Re_m^{-1}\|\nabla\varphi\|_0^2 \end{aligned}$$

$$
\begin{aligned}
&\geq \beta_1 Re^{-1}\|\vec{v}\|_1^2 + \xi Re_m^{-1}\|\varphi\|_1^2 \\
&\geq \widetilde{c}_\alpha \min\{Re^{-1}, Re_m^{-1}\}\|(\vec{v},\varphi)\|_1^2,
\end{aligned}
$$

where $\widetilde{c}_\alpha = \min\{\beta_1, \xi\}$, β_1 comes from Korn's Inequality, and ξ is the constant from Friedrichs' inequality [5],

$$
\|\nabla\varphi\|_0^2 \geq \xi\|\varphi\|_1^2, \quad \forall \varphi \in H_\tau^1(\Omega).
$$

Now, we prove the second inequality,

$$
\begin{aligned}
\widetilde{\mathcal{A}}(\vec{u},\psi;\vec{v},\varphi) &= Re^{-1}\int_\Omega \mathcal{S}\vec{u} : \mathcal{S}\vec{v}\,\mathrm{dX} + Re_m^{-1}\int_\Omega \nabla\psi\cdot\nabla\varphi\,\mathrm{dX} \\
&\leq 2Re^{-1}\|\vec{u}\|_1\|\vec{v}\|_1 + Re_m^{-1}\|\psi\|_1\|\varphi\|_1 \\
&\leq \max\{2Re^{-1}, Re_m^{-1}\}\|(\vec{u},\psi)\|_1\|(\vec{v},\varphi)\|_1.
\end{aligned}
$$

The last inequality is obtained by the Cauchy-Schwarz inequality.

□

3.2 Mixed variational formulation

Noting that $b(q,\vec{u}) = 0$ for all $q \in \mathbf{Q}$, since $\vec{u} \in \mathbf{J}$, the mixed variational formulation (21)-(23) is equivalent to,

Formulation 4. *Find* $(\vec{u}, p, A) \in \mathbf{J}\times\mathbf{Q}\times\widetilde{\mathbf{X}}$ *such that*

$$
\begin{aligned}
a_1(\vec{u},\vec{v}) + c_0(\vec{u};\vec{u},\vec{v}) + \widetilde{c}_1(A;\vec{v},A) + b(p,\vec{v}) &= \langle \vec{f},\vec{v}\rangle, \\
\widetilde{a}_2(A,\psi) + \widetilde{c}_2(A;\vec{u},\psi) &= \langle -E^0,\psi\rangle,
\end{aligned}
$$

for all $(\vec{v},\psi) \in \mathbf{W}\times\widetilde{\mathbf{X}}$.

We introduce the extended form

$$
\widetilde{\mathcal{B}}(q;\vec{v},\psi) := b(q,\vec{v}).
$$

The difference between $\mathcal{B}$ and $\widetilde{\mathcal{B}}$ is that they act on $\mathbf{X}$ and $\widetilde{\mathbf{X}}$, respectively.

Formulation 4 can be written in a compact form:

Formulation 5. *Find* $(\vec{u}, p, A) \in \mathbf{W}\times\mathbf{Q}\times\widetilde{\mathbf{X}}$ *such that*

$$
\begin{aligned}
\widetilde{\mathcal{A}}(\vec{u},A;\vec{v},\psi) + \widetilde{\mathcal{C}}(\vec{u},A;\vec{u},A;\vec{v},\psi) + \widetilde{\mathcal{B}}(p;\vec{v},\psi) &= \widetilde{\mathcal{L}}(\vec{v},\psi), \\
\widetilde{\mathcal{B}}(q;\vec{u},A) &= 0,
\end{aligned}
\tag{24}
$$

for all $(\vec{v},q,\psi) \in \mathbf{W}\times\mathbf{Q}\times\widetilde{\mathbf{X}}$.

The form $\widetilde{\mathcal{B}}$ has the following property

$$
|\widetilde{\mathcal{B}}(q;\vec{v},\psi)| \leq C_b\|q\|_0\|\vec{v}\|_1 \leq \widetilde{C}_b\|q\|_0\|(\vec{v},\psi)\|_1, \tag{25}
$$

for all $(\vec{v},q,\psi) \in \mathbf{W}\times\mathbf{Q}\times\widetilde{\mathbf{X}}$, with a constant $\widetilde{C}_b > 0$. Furthermore, we have the inf-sup condition for the form $\widetilde{\mathcal{B}}$:

Lemma 9. *There exists a constant $\Gamma > 0$ depending only on Ω such that*

$$\sup_{(\vec{0},0)\neq(\vec{v},\psi)\in\mathbf{W}\times\widetilde{\mathbf{X}}} \frac{\widetilde{\mathcal{B}}(q;\vec{v},\psi)}{\|(\vec{v},\psi)\|_1} \geq \Gamma\|q\|_0, \tag{26}$$

for all $q \in \mathbf{Q}$.

Proof: The proof is similar to Lemma 6, so we omit it. □

3.3 Relationship between solutions of the two formulations

For the steady-state visco-resistive MHD problem, (1)-(4), we establish two weak formulations in Sections 2 and 3. A natural question is whether the solutions of these two formulations are the same. Here, we provide conditions under which this is the case. These results follow naturally from the fact that $\mathbf{X} \subseteq \widetilde{\mathbf{X}}$.

Theorem 4. *Assume that $(\vec{u}, p, A) \in \mathbf{W} \times \mathbf{Q} \times \mathbf{X}$ is a solution of Formulation 2, then $(\vec{u}, p, A)$ is also a solution of Formulation 5.*

Proof: Let $(\vec{u}, p, A) \in \mathbf{W}\times\mathbf{Q}\times\mathbf{X}$ be a solution of Formulation 2. According to (7), the following equality holds

$$\int_\Omega \triangle A \cdot (\nabla A \cdot \vec{v})\,\mathrm{dX} = -\int_\Omega (\nabla \cdot T_M) \cdot \vec{v}\,\mathrm{dX} = \int_\Omega T_M : \nabla\vec{v}\,\mathrm{dX}, \quad \forall \vec{v} \in \mathbf{W}.$$

Then, (8) is the same as (21). For any $\psi \in H^1_\tau(\Omega) \subseteq L^2(\Omega)$, there exists $\varphi \in H^2_\tau(\Omega)$ such that $\triangle\varphi = \psi$ (see [12, Chapter I, Theorem 1.10]). In (9),

$$\int_\Omega -\vec{u} \cdot \nabla A \cdot \triangle\varphi\,\mathrm{dX} + \int_\Omega Re_m^{-1} \triangle A \cdot \triangle\varphi\,\mathrm{dX} = \langle E^0, \triangle\varphi\rangle, \quad \forall \varphi \in \mathbf{X},$$

taking $\triangle\varphi = \psi$ implies (22). So $(\vec{u}, p, A)$ is also a solution of Formulation 22. □

Theorem 5. *Assume that $(\vec{u}, p, A) \in \mathbf{W} \times \mathbf{Q} \times \widetilde{\mathbf{X}}$ is a solution of Formulation 5 and that this solution is smooth enough such that $A \in \mathbf{X}$. Then, $(\vec{u}, p, A)$ is also a solution of Formulation 2.*

Proof: Let $(\vec{u}, p, A) \in \mathbf{W} \times \mathbf{Q} \times \widetilde{\mathbf{X}}$ be a solution of Formulation 5. Since $A \in \mathbf{X}$, the following equality holds

$$\int_\Omega T_M : \nabla\vec{v}\,\mathrm{dX} = -\int_\Omega (\nabla \cdot T_M) \cdot \vec{v}\,\mathrm{dX} = \int_\Omega \triangle A \cdot (\nabla A \cdot \vec{v})\,\mathrm{dX}, \quad \forall \vec{v} \in \mathbf{W}.$$

Then, (21) is the same as (8). Furthermore,

$$\int_\Omega \left[\vec{u} \cdot \nabla A \cdot \psi + Re_m^{-1} \nabla A \cdot \nabla\psi\right]\mathrm{dX} = -\int_\Omega E^0 \cdot \psi\,\mathrm{dX}, \quad \forall \psi \in \widetilde{\mathbf{X}},$$

can be rewritten as

$$\int_\Omega \nabla A \cdot \nabla \psi \mathrm{dX} = -Re_m \int_\Omega \left(E^0 + \vec{u} \cdot \nabla A\right) \cdot \psi \,\mathrm{dX}, \quad \forall \psi \in \widetilde{\mathbf{X}}.$$

Since $\int_\Omega E^0 \mathrm{dX} = 0$ and $\int_\Omega \vec{u} \cdot \nabla A \,\mathrm{dX} = -\int_\Omega (\nabla \cdot \vec{u}) A \,\mathrm{dX} + \int_{\partial\Omega} (\vec{u} \cdot \vec{n}) A \,\mathrm{dX} = 0$, we have $\int_\Omega (E^0 + \vec{u} \cdot \nabla A) \,\mathrm{dX} = 0$. Using the results of Proposition 1.2 of [12], shows that the weak form of finding $w \in H^1_\tau(\Omega)$ such that

$$\int_\Omega \nabla w \cdot \nabla \psi \mathrm{dX} = \int_\Omega -Re_m \left(E^0 + \vec{u} \cdot \nabla A\right) \cdot \psi \,\mathrm{dX}, \quad \forall \psi \in \widetilde{\mathbf{X}},$$

has a unique solution, and that if $w \in H^2_\tau(\Omega)$, then it is the strong solution of the Neumann problem,

$$\left\{ \begin{array}{rcl} -\Delta w & = & -Re_m(E^0 + \vec{u} \cdot \nabla A), \quad \text{in } \Omega, \\ \frac{\partial w}{\partial \vec{n}} & = & 0, \quad \text{on } \partial\Omega, \\ \int_\Omega w \,\mathrm{dX} & = & 0. \end{array} \right. \tag{27}$$

Thus, from [12, Chapter I, Theorem 1.10], we have that (27) has a unique solution, $w \in H^2_\tau(\Omega)$, which is given by $w = A$, implying that $-\vec{u} \cdot \nabla A + Re_m^{-1} \triangle A = E^0$. For $\varphi \in H^2_\tau(\Omega)$, multiplying both sides by $\triangle \varphi$ and integrating yields (9). So $(\vec{u}, p, A)$ is also a solution of Formulation 2. □

Theorem 6. *Assume that (20) holds. Then, Formulation 5 has at least one solution $(\vec{u}, p, A) \in \mathbf{W} \times \mathbf{Q} \times \widetilde{\mathbf{X}}$, which is the unique solution of Formulation 2. Furthermore, if all of the solutions of Formulation 5 satisfy $(\vec{u}, p, A) \in \mathbf{W} \times \mathbf{Q} \times \mathbf{X}$, then Formulation 2 and Formulation 5 have the same solution, and the solution is unique.*

Proof: Since (20) holds, Theorem 3 states that Formulation 2 has a unique solution $(\vec{u}, p, A)$. According to Theorem 4, $(\vec{u}, p, A)$ is also a solution of Formulation 5.

If $A \in \mathbf{X}$, Theorem 5 states that the solution $(\vec{u}, p, A)$ of Formulation 5 is also a solution of Formulation 2. However, under the condition of (20), Formulation 2 has only one solution. This means that Formulation 5 has only one solution. □ The above theorems establish that, under some conditions, both the "curled" and "uncurled" formulations are not only well posed, but also have the same solutions. In the following, we consider the properties of the solutions of discretizations of the uncurled formulation.

3.4 Finite element discretization

In this subsection, we introduce a mixed finite-element approximation of the uncurled formulation and discuss the convergence rates that are obtained under some standard smoothness assumptions.

Before discussing the finite-element approximation, we introduce the finite-element space. Let $\mathcal{T}_h$ be a quasi-uniform family of subdivisions that partition Ω

into triangles or quadrilaterals, $\mathcal{K}$, with diameters bounded by h [12, Chapter I, Definitions A.2]. Based on these meshes, we construct a series of finite-element spaces satisfying

$$\mathbf{W}_h \subset \mathbf{W}, \mathbf{X}_h \subset \widetilde{\mathbf{X}}, \mathbf{Q}_h \subset \mathbf{Q},$$

as well as the Kernel space

$$\mathbf{Z}_h := \{\vec{v} \in \mathbf{W}_h | b(q, \vec{v}) = 0, \forall q \in \mathbf{Q}_h\}.$$

The discrete problem of Formulation 5 can be written as

Formulation 6. *Find* $(\vec{u}_h, p_h, A_h) \in \mathbf{W}_h \times \mathbf{Q}_h \times \mathbf{X}_h$ *such that*

$$\begin{aligned} \widetilde{\mathcal{A}}(\vec{u}_h, A_h; \vec{v}, \psi) + \widetilde{\mathcal{C}}(\vec{u}_h, A_h; \vec{u}_h, A_h; \vec{v}, \psi) + \widetilde{\mathcal{B}}(p_h; \vec{v}, \psi) &= \widetilde{\mathcal{L}}(\vec{v}, \psi), \\ \widetilde{\mathcal{B}}(q; \vec{u}_h, A_h) &= 0, \end{aligned}$$

for all $(\vec{v}, q, \psi) \in \mathbf{W}_h \times \mathbf{Q}_h \times \mathbf{X}_h$.

Some standard properties of Sobolev spaces are useful in the analysis to follow.

Theorem 7 ([1], Theorem IV4.12)**.** *Let* Ω *be a bounded and open Lipschitz-continuous subset of* $\mathbb{R}^n$*. If* $mp > n$*, then we have the imbedding*

$$W^{m,p}(\Omega) \hookrightarrow L^q(\Omega), \quad \text{for} \quad p \leq q \leq \infty.$$

In this paper, we consider the 2D problem and assume that the solution $A \in H^{s+1}(\Omega), s \geq 1$, then from Theorem 7 we have

$$|\nabla A|_\infty \leq C_A \|\nabla A\|_s \leq C_A \|A\|_{s+1}, \quad s \geq 1. \tag{28}$$

Theorem 8. *Assume that* $(\vec{u}, A)$ *is the solution of Formulation 5 with* $\vec{u} \in \left(H^1(\Omega)\right)^2$ *and* $A \in H^{s+1}(\Omega)$ *for* $s \geq 1$*, and* $(\vec{u}_h, A_h)$ *is the solution of Formulation 6 satisfying* $\|\vec{u}_h\|_1 + |\nabla A_h|_\infty \leq d$*, where* d *is a constant. Then, we have the following estimate*

$$\|(\vec{u} - \vec{u}_h, A - A_h)\|_1 \leq C\big(\inf_{(\vec{v},\psi)\in \mathbf{W}_h \times \mathbf{X}_h} \|(\vec{u} - \vec{v}, A - \psi)\|_1 + \inf_{q \in \mathbf{Q}_h} \|p - q\|_0\big),$$

with a constant $C > 0$*, for sufficiently small values of* Re*,* Re_m*.*

Proof: Subtracting Formulation 6 from Equality (24), we have

$$\begin{aligned} &\widetilde{\mathcal{A}}(\vec{u} - \vec{u}_h, A - A_h; \vec{v}, \psi) + \widetilde{\mathcal{C}}(\vec{u} - \vec{u}_h, A - A_h; \vec{u}, A; \vec{v}, \psi) \\ &\quad + \widetilde{\mathcal{C}}(\vec{u}_h, A_h; \vec{u} - \vec{u}_h, A - A_h; \vec{v}, \psi) + \widetilde{\mathcal{B}}(p - p_h; \vec{v}, \psi) = 0, \end{aligned} \tag{29}$$

for all $(\vec{v}, \psi) \in \mathbf{W}_h \times \mathbf{X}_h$.

Taking $\vec{v} \in \mathbf{Z}_h$ and using (29), we have

$$\widetilde{\mathcal{A}}(\vec{v} - \vec{u}_h, \psi - A_h; \vec{v} - \vec{u}_h, \psi - A_h)$$

$$
\begin{aligned}
& +\widetilde{\mathcal{C}}(\vec{v}-\vec{u}_h,\psi-A_h;\vec{u},A;\vec{v}-\vec{u}_h,\psi-A_h) \\
& +\widetilde{\mathcal{C}}(\vec{u}_h,A_h;\vec{v}-\vec{u}_h,\psi-A_h;\vec{v}-\vec{u}_h,\psi-A_h) \\
= \quad & \widetilde{\mathcal{A}}(\vec{v}-\vec{u},\psi-A;\vec{v}-\vec{u}_h,\psi-A_h) \\
& +\widetilde{\mathcal{C}}(\vec{v}-\vec{u},\psi-A;\vec{u},A;\vec{v}-\vec{u}_h,\psi-A_h) \\
& +\widetilde{\mathcal{C}}(\vec{u}_h,A_h;\vec{v}-\vec{u},\psi-A;\vec{v}-\vec{u}_h,\psi-A_h) \\
& -\widetilde{\mathcal{B}}(p-p_h;\vec{v}-\vec{u}_h,\psi-A_h), \qquad (30)
\end{aligned}
$$

Since $\vec{v}\in\mathbf{Z}_h$, we have

$$
\widetilde{\mathcal{B}}(p-p_h;\vec{v}-\vec{u}_h,\psi-A_h)=\widetilde{\mathcal{B}}(p-q;\vec{v}-\vec{u}_h,\psi-A_h), \qquad (31)
$$

for all $q\in\mathbf{Q}_h$.

From (30) and (31), we have the estimate

$$
\begin{aligned}
\text{r.h.s of (30)} \quad \leq \quad & \|(\vec{v}-\vec{u}_h,\psi-A_h)\|_1\big[\max\{2Re^{-1},Re_m^{-1}\}\|(\vec{v}-\vec{u},\psi-A)\|_1 \\
& +C\|(\vec{v}-\vec{u},\psi-A)\|_1\big(\|\vec{u}\|_1+C_A\|A\|_{s+1,2}\big) \\
& +C\|(\vec{v}-\vec{u},\psi-A)\|_1\big(\|\vec{u}_h\|_1+|\nabla A_h|_\infty\big)+\widetilde{C}_b\|p-q\|_0\big] \\
\leq \quad & C_r\|(\vec{v}-\vec{u}_h,\psi-A_h)\|_1\big(\|(\vec{u}-\vec{v},A-\psi)\|_1+\|p-q\|_0\big), \qquad (32)
\end{aligned}
$$

where $C_r=\max\{2Re^{-1},Re_m^{-1}\}+2C\cdot\max\{\|\vec{u}\|_1+C_A\|A\|_{s+1,2},\|\vec{u}_h\|_1+|\nabla A_h|_\infty\}+\widetilde{C}_b$, C_A comes from (28), and $\widetilde{C}_b$ comes from (25). Since $(\vec{u},A)$ is the solution of the continuous problem and $\vec{u}\in H^1(\Omega)$ and $A\in H^{s+1}(\Omega)$, then $\|\vec{u}\|_1+C_A\|A\|_{s+1,2}$ can be bounded by some constant. By assumption, so can $\|\vec{u}_h\|_1+|\nabla A_h|_\infty$.

Similarly,

$$
\begin{aligned}
\text{l.h.s of (30)} \quad \geq \quad & \widetilde{c}_\alpha\min\{Re^{-1},Re_m^{-1}\}\cdot\|(\vec{v}-\vec{u}_h,\psi-A_h)\|_1^2 \\
& -C\|(\vec{v}-\vec{u}_h,\psi-A_h)\|_1^2\cdot\big(\|\vec{u}\|_1+\|A\|_{s+1,2}\big) \\
& -C\|(\vec{v}-\vec{u}_h,\psi-A_h)\|_1^2\cdot\big(\|\vec{u}_h\|_1+|\nabla A_h|_\infty\big) \\
\geq \quad & C_l\|(\vec{v}-\vec{u}_h,\psi-A_h)\|_1^2, \qquad (33)
\end{aligned}
$$

where $C_l=\widetilde{c}_\alpha\min\{Re^{-1},Re_m^{-1}\}-2C\cdot\max\{\|\vec{u}\|_1+C_A\|A\|_{s+1,2},\|\vec{u}_h\|_1+|\nabla A_h|_\infty\}$ and $\widetilde{c}_\alpha$ comes from Lemma 8. Here, we assume that $\widetilde{c}_\alpha\min\{Re^{-1},Re_m^{-1}\}$ is large enough such that $C_l\geq\dfrac{\widetilde{c}_\alpha}{2}\min\{Re^{-1},Re_m^{-1}\}$.

According to (32) and (33), we have the following estimate

$$
\|(\vec{v}-\vec{u}_h,\psi-A_h)\|_1\leq C\bigg(\|(\vec{u}-\vec{v},A-\psi)\|_1+\|p-q\|_0\bigg),
$$

where $C=C_r/C_l$. Furthermore,

$$
\begin{aligned}
\|(\vec{u}-\vec{u}_h,A-A_h)\|_1 \quad \leq \quad & \sqrt{2}\big(\|(\vec{u}-\vec{v},A-\psi)\|_1+\|(\vec{v}-\vec{u}_h,\psi-A_h)\|_1\big), \\
\leq \quad & C\|(\vec{u}-\vec{v},A-\psi)\|_1+C\|p-q\|_0.
\end{aligned}
$$

Let $\vec{v} \in W_h$ be arbitrary and $\vec{w} \in W_h$ be a solution of

$$b(q, \vec{w}) = b(q, \vec{u} - \vec{v}), \quad \forall q \in \mathbf{Q}_h.$$

Since b satisfies an inf-sup condition and a continuity condition, then there exists a solution to this problem such that

$$\|\vec{w}\|_1 \leq C\|\vec{u} - \vec{v}\|_1,$$

and $(\vec{w} + \vec{v}) \in \mathbf{Z}_h$. By the triangle inequality and using the result above, we have

$$\begin{aligned} \|(\vec{u} - \vec{u}_h, A - A_h)\|_1 &\leq C\|(\vec{u} - (\vec{w} + \vec{v}), A - \psi)\|_1 + C\|p - q\|_0, \\ &\leq C\|(\vec{u} - \vec{v}, A - \psi)\|_1 + C\|\vec{w}\|_1 + C\|p - q\|_0, \\ &\leq C\|(\vec{u} - \vec{v}, A - \psi)\|_1 + C\|p - q\|_0. \end{aligned}$$

□ To give a more precise definition of our finite-element approximations, we need a few definitions. Define on an element $\mathcal{K}$,

$$\mathcal{P}_k(\mathcal{K}) := \text{the space of polynomials of degree} \leq k.$$

Let

$$\begin{aligned} \mathbf{W_h} &:= \{\vec{v}_h : \vec{v}_h|_{\mathcal{K}} \in (\mathcal{P}_{k+1})^2, \quad \forall \mathcal{K} \in \mathcal{T}_h\}, \\ \mathbf{Q_h} &:= \{q_h : q_h|_{\mathcal{K}} \in \mathcal{P}_k, \quad \forall \mathcal{K} \in \mathcal{T}_h\}, \\ \mathbf{X_h} &:= \{\psi_h : \psi_h|_{\mathcal{K}} \in \mathcal{P}_{k+1}, \quad \forall \mathcal{K} \in \mathcal{T}_h\}. \end{aligned}$$

Assumption 1. *Let $k \geq 1, s \geq 1$. Based on the above finite-element spaces, the following approximation properties hold*

$$\inf_{\vec{v}_h \in \mathbf{W}_h} \|\vec{u} - \vec{v}_h\|_1 + \inf_{q_h \in \mathbf{Q}_h} \|p - q_h\|_0 \leq Ch^{\min\{s,k\}} \left[\|u\|_{s+1} + \|p\|_s\right],$$

for all $(\vec{u}, p) \in H^{s+1}(\Omega)^2 \times H^s(\Omega)$ and

$$\inf_{\psi_h \in \mathbf{X}_h} \|A - \psi_h\|_1 \leq Ch^{\min\{s,k\}} \|A\|_{s+1},$$

for all $A \in H^{s+1}(\Omega)$.

Theorem 9. *Let the solution, $(\vec{u}, A)$, of Formulation 5 satisfy*

$$\vec{u} \in \left(H^{s+1}(\Omega)\right)^2, \quad A \in H^{s+1}(\Omega), \quad p \in H^s(\Omega)$$

and Assumption 1 hold. Let $(\vec{u}_h, A_h) \in \mathbf{W}_h \times \mathbf{X}_h$ be the finite-element approximation in Formulation 6. Then, we have the error bound

$$\|(\vec{u} - \vec{u}_h, A - A_h)\|_1 \leq Ch^{\min\{s,k\}} \left[\|\vec{u}\|_{s+1} + \|p\|_s + \|A\|_{s+1}\right].$$

Now we have established the estimate of the finite-element approximation to the continuum problem. In the following section, we will give the convergence analysis of Newton's method to the discrete finite-element problem. Finally, we obtain the convergence analysis of the continuum problem by Newton's method.

4 Newton's Method

Since the weak formulation in (21)-(23) is nonlinear, we use Newton's method to derive a linearized system. As expected, the discrete form leads to a saddle-point problem [3, 6]. Here, we focus on the linearization steps and show that the resulting systems are well-posed, and that the solutions converge to that of the original problem, under certain assumptions.

4.1 Newton linearizations

Let $\mathbf{S} = \left(H_0^1(\Omega)\right)^2 \times H_\tau^1(\Omega)$ with the norm $\|W\|_1^2 = \|\vec{v}\|_1^2 + \|\psi\|_1^2$ for all $W = (\vec{v}, \psi) \in \mathbf{S}$. For convenience, we denote the solutions of Formulation 5 and Formulation 6 as $(U^*, p^*), (U_h^*, p_h^*)$, respectively.

For $U = (\vec{u}, A), W = (\vec{v}, \psi) \in \mathbf{S}$, define the following operators:

$$\begin{aligned}
\mathcal{L}_1(\vec{u}, A, p)[\vec{v}] &:= a_1(\vec{u}, \vec{v}) + b(p, \vec{v}) + c_0(\vec{u}; \vec{u}, \vec{v}) + \widetilde{c}_1(A; \vec{v}, A) - \langle \vec{f}, \vec{v} \rangle, \\
\mathcal{L}_2(\vec{u}, A, p)[\psi] &:= \widetilde{a}_2(A, \psi) + \widetilde{c}_2(A; \vec{u}, \psi) + \langle E^0, \psi \rangle, \\
\mathcal{L}_3(\vec{u}, A, p)[q] &:= -b(q, \vec{u}).
\end{aligned}$$

Problem (21)-(23) is equivalent to

$$\mathcal{L}_1(\vec{u}, A, p)[\vec{v}] = 0, \qquad \forall \vec{v} \in \mathbf{W}, \tag{34}$$

$$\mathcal{L}_2(\vec{u}, A, p)[\psi] = 0, \qquad \forall \psi \in \widetilde{\mathbf{X}}, \tag{35}$$

$$\mathcal{L}_3(\vec{u}, A, p)[q] = 0, \qquad \forall q \in \mathbf{Q}.$$

Since the variational system contains nonlinearities in both (34) and (35), we linearize the above forms. Let $\vec{u}_k, A_k, p_k$ be the current approximations for $\vec{u}, A, p$, respectively and $\delta\vec{u}_k = \vec{u}_{k+1} - \vec{u}_k, \delta A = A_{k+1} - A_k, \delta p = p_{k+1} - p_k$ be the update to the approximations, then the linear systems that arise within Newton iterations are denoted

$$\begin{bmatrix} \mathcal{L}_{1,\vec{u}} & \mathcal{L}_{1,A} & \mathcal{L}_{1,p} \\ \mathcal{L}_{2,\vec{u}} & \mathcal{L}_{2,A} & 0 \\ \mathcal{L}_{3,\vec{u}} & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta\vec{u} \\ \delta A \\ \delta p \end{bmatrix} = - \begin{bmatrix} \mathcal{L}_1 \\ \mathcal{L}_2 \\ \mathcal{L}_3 \end{bmatrix},$$

where each of the system components is evaluated at $\vec{u}_k, A_k, p_k$. That is

$$\begin{aligned}
\mathcal{L}_{1,\vec{u}}[\vec{v}] \cdot \delta\vec{u} &= \frac{\partial}{\partial \vec{u}}(\mathcal{L}_1(\vec{u}_k, A_k, p_k)[\vec{v}])[\delta\vec{u}], \\
\mathcal{L}_{1,A}[\vec{v}] \cdot \delta A &= \frac{\partial}{\partial A}(\mathcal{L}_1(\vec{u}_k, A_k, p_k)[\vec{v}])[\delta A], \\
\mathcal{L}_{1,p}[\vec{v}] \cdot \delta p &= \frac{\partial}{\partial p}(\mathcal{L}_1(\vec{u}_k, A_k, p_k)[\vec{v}])[\delta p], \\
\mathcal{L}_{2,\vec{u}}[\psi] \cdot \delta\vec{u} &= \frac{\partial}{\partial \vec{u}}(\mathcal{L}_2(\vec{u}_k, A_k, p_k)[\psi])[\delta\vec{u}], \\
\mathcal{L}_{2,A}[\psi] \cdot \delta A &= \frac{\partial}{\partial A}(\mathcal{L}_2(\vec{u}_k, A_k, p_k)[\psi])[\delta A],
\end{aligned}$$

$$\mathcal{L}_{3,\vec{u}}[q]\cdot\delta\vec{u} \quad = \quad \frac{\partial}{\partial\vec{u}}(\mathcal{L}_3(\vec{u}_k, A_k, p_k)[q])[\delta\vec{u}].$$

Computing the Gâteaux derivatives yields

$$\begin{aligned}
\mathcal{L}_{1,\vec{u}}[\vec{v}]\cdot\delta\vec{u} &= a_1(\delta\vec{u},\vec{v}) + c_0(\vec{u}_k;\delta\vec{u},\vec{v}) + c_0(\delta\vec{u};\vec{u}_k,\vec{v}),\\
\mathcal{L}_{1,A}[\vec{v}]\cdot\delta A &= \hat{a}(A_k;\vec{v},\delta A),\\
\mathcal{L}_{1,p}[\vec{v}]\cdot\delta p &= b(\delta p,\vec{v}),\\
\mathcal{L}_{2,\vec{u}}[\psi]\cdot\delta\vec{u} &= \widetilde{c}_2(A_k;\delta\vec{u},\psi),\\
\mathcal{L}_{2,A}[\psi]\cdot\delta A &= \widetilde{a}_2(\delta A,\psi) + \widetilde{c}_2(\delta A;\vec{u}_k,\psi),\\
\mathcal{L}_{3,\vec{u}}[q]\cdot\delta\vec{u} &= b(p_k,\delta\vec{u}),
\end{aligned}$$

where

$$\begin{aligned}
\hat{a}(A_k;\vec{v},A) \quad := \quad & \left\langle \left(\frac{\partial A_k}{\partial y}\cdot\frac{\partial A}{\partial y} - \frac{\partial A_k}{\partial x}\cdot\frac{\partial A}{\partial x}, -\left[\frac{\partial A_k}{\partial x}\cdot\frac{\partial A}{\partial y} + \frac{\partial A}{\partial x}\cdot\frac{\partial A_k}{\partial y}\right]\right), \frac{\partial\vec{v}}{\partial x}\right\rangle_0\\
& + \left\langle \left(-\left[\frac{\partial A_k}{\partial x}\cdot\frac{\partial A}{\partial y} + \frac{\partial A}{\partial x}\cdot\frac{\partial A_k}{\partial y}\right], \frac{\partial A_k}{\partial x}\cdot\frac{\partial A}{\partial x} - \frac{\partial A_k}{\partial y}\cdot\frac{\partial A}{\partial y}\right), \frac{\partial\vec{v}}{\partial y}\right\rangle_0 .
\end{aligned}$$

Define the following forms:

$$\begin{aligned}
\mathfrak{A}(U_k;U,W) \quad := \quad & \hat{a}(A_k;\vec{v},A) + a_1(\vec{u},\vec{v}) + \widetilde{a}_2(A,\psi) + c_0(\vec{u}_k;\vec{u},\vec{v}) + c_0(\vec{u};\vec{u}_k,\vec{v})\\
& + \widetilde{c}_2(A_k;\vec{u},\psi) + \widetilde{c}_2(A;\vec{u}_k,\psi),\\
\mathfrak{B}(W,q) \quad := \quad & b(q,\vec{v}).
\end{aligned}$$

Using Newton's method to get an approximation to Formulation 5 yields the Newton system,

Formulation 7. *Find* $(\delta U,\delta p)\in\mathbf{S}\times\mathbf{Q}$ *such that*

$$\begin{aligned}
\mathfrak{A}(U_k;\delta U,W) + \mathfrak{B}(W,\delta p) &= F(U_k,p_k;W),\\
\mathfrak{B}(\delta U,q) &= G(U_k;q),
\end{aligned}$$

for all $(W,q)\in\mathbf{S}\times\mathbf{Q}$*, where*

$$\begin{aligned}
F(U_k,p_k;W) &= \widetilde{\mathcal{L}}(\vec{v},\psi) - \widetilde{\mathcal{A}}(\vec{u}_k,A_k;\vec{v},\psi)\\
&\quad - \widetilde{\mathcal{C}}(\vec{u}_k,A_k;\vec{u}_k,A_k;\vec{v},\psi) - \widetilde{\mathcal{B}}(p_k;\vec{v},\psi)\\
G(U_k;q) &= -\mathfrak{B}(U_k,q).
\end{aligned}$$

At the continuum level, existence and uniqueness of solutions to these formulations is difficult with $A\in H^1(\Omega)$ instead of $H^2(\Omega)$. Thus, we now focus our analysis on the discrete case.

4.2 Discrete system

Denote $C_{sup} = \sup_{(x,y)\in\Omega} |\nabla \vec{u}_k|, D_{sup} = \sup_{(x,y)\in\Omega} |\nabla A_k|$, and $M_{sup} = \sup_{(x,y)\in\Omega} |\vec{u}_k|$, and assume they are all finite, as they will be for finite-element approximations A_k, and $\vec{u}_k$.

Lemma 10. *$\mathfrak{A}(U_k; U, W)$ and $\mathfrak{B}(W, q)$ are continuous for the norms $\|\cdot\|_1$ and $\|\cdot\|_0$.*

Proof: For the continuity of $\mathfrak{A}(U_k; U, W)$, observe that

$$\begin{aligned} |\mathfrak{A}(U_k; U, W)| \quad \leq \quad & |\hat{a}(A_k; \vec{v}, A) + a_1(\vec{u}, \vec{v}) + \widetilde{a}_2(A, \psi) + c_0(\vec{u}_k; \vec{u}, \vec{v}) \\ & + c_0(\vec{u}; \vec{u}_k, \vec{v}) + \widetilde{c}_2(A_k; \vec{u}, \psi) + \widetilde{c}_2(A; \vec{u}_k, \psi)|. \end{aligned}$$

Next, consider the above summands separately. First, note that

$$|\hat{a}(A_k; \vec{v}, A)| \leq 2D_{sup} \|\nabla A\|_0 \|\nabla \vec{v}\|_0.$$

Recalling the definitions of the rest of these terms, we can obtain the following estimates

$$\begin{aligned} |a_1(\vec{u}, \vec{v})| &\leq CR_e^{-1} \|\vec{u}\|_1 \|\vec{v}\|_1, \\ |\widetilde{a}_2(A, \psi)| &\leq Re_m^{-1} \|A\|_1 \|\psi\|_1, \\ |c_0(\vec{u}_k; \vec{u}, \vec{v})| &\leq M_{sup} \|\nabla \vec{u}\|_0 \|\vec{v}\|_0, \\ |c_0(\vec{u}; \vec{u}_k, \vec{v})| &\leq C_{sup} \|\vec{u}\|_0 \|\vec{v}\|_0, \\ |\widetilde{c}_2(A_k; \vec{u}, \psi)| &\leq D_{sup} \|\vec{u}\|_0 \|\psi\|_0, \\ |\widetilde{c}_2(A; \vec{u}_k, \psi)| &\leq M_{sup} \|\nabla A\|_0 \|\psi\|_0, \end{aligned}$$

An application of the Cauchy-Schwarz inequality shows that

$$|\mathfrak{A}(U_k; U, W)| \leq C \|U\|_1 \|W\|_1,$$

where C is a constant.

For the continuity of $\mathfrak{B}(W, q)$, by Hölder's inequality, we have

$$\begin{aligned} |\mathfrak{B}(W, q)| = |-b(q, \vec{v})| \quad &= \quad \left| \int_\Omega q(\nabla \cdot \vec{v}) \, \mathrm{dX} \right| \\ &\leq \quad \|q\|_0 \|\nabla \cdot \vec{v}\|_0 \\ &\leq \quad \|q\|_0 \|\vec{v}\|_1 \\ &\leq \quad \|q\|_0 \|W\|_1. \end{aligned}$$

Therefore, $\mathfrak{B}(W, q)$ is a continuous bilinear form. □

Lemma 11. *$F(U_k, p_k; W)$ and $G(U_k; q)$ are bounded linear functionals on $\mathbf{S}$ and $\mathbf{Q}$, respectively.*

Proof: For simplicity, we consider the components of $F(U_k, p_k; W)$. Applying the Cauchy-Schwarz inequality yields

$$\begin{aligned}
|\hat{a}(A_k; \vec{v}, A_k)| &\leq CD_{sup}\|\nabla A_k\|_0\|\nabla \vec{v}\|_0, \\
|a_1(\vec{u}_k, \vec{v})| &\leq CRe^{-1}\|\vec{u}_k\|_1\|\vec{v}\|_1, \\
|\widetilde{a}_2(A_k, \psi)| &\leq Re_m^{-1}\|A_k\|_1\|\psi\|_1, \\
|c_0(\vec{u}_k; \vec{u}_k, \vec{v})| &\leq C_{sup}\|\vec{u}_k\|_0\|\psi\|_0, \\
|\widetilde{c}_2(A_k; \vec{u}_k, \psi)| &\leq D_{sup}\|\vec{u}_k\|_0\|\psi\|_0, \\
|\langle E^0, \psi\rangle_0| &\leq \|E^0\|_0\|\psi\|_0, \\
|\langle \vec{f}, \vec{v}\rangle| &\leq \|\vec{f}\|_{-1}\|\vec{v}\|_1,
\end{aligned}$$

where C is a constant.

Combining the above inequalities and the continuity of $b(q, \vec{v})$, we have

$$|F(U_k, p_k; W)| \leq C\|W\|_1,$$

where C is a constant only depending on different norms of U_k.

By Hölder's inequality, we have

$$|G(U_k; q)| = |-\mathfrak{B}(U_k, q)| \leq \|U_k\|_1\|q\|_0,$$

implying that $G(U_k; q)$ is bounded. □

Next, we prove the existence and uniqueness of the solutions to the discrete Newton systems at each step. Under certain conditions, we prove that $\mathfrak{A}(U_k; U, W)$ is coercive. We use the finite-element spaces $\mathbf{S}_h := \mathbf{W}_h \times \mathbf{X}_h \subset \mathbf{S}$ and $\mathbf{Q}_h \subset \mathbf{Q}$, and the discrete Newton system of Formulation 7 is

Formulation 8. *Find* $(\delta U_h, \delta p_h) \in \mathbf{S}_h \times \mathbf{Q}_h$ *such that*

$$\mathfrak{A}(U_{h,k}; \delta U_h, W_h) + \mathfrak{B}(W_h, \delta p_h) = F(U_{h,k}, p_{h,k}; W_h), \tag{36}$$

$$\mathfrak{B}(\delta U_h, q_h) = G(U_{h,k}; q_h), \tag{37}$$

for all $(W_h, q_h) \in \mathbf{S}_h \times \mathbf{Q}_h$. *Let* $U_{h,k+1} = U_{h,k} + \delta U_h, p_{h,k+1} = p_{h,k} + \delta p_h$.

For simplicity, throughout the remainder of this section, we drop the subscript h. To illustrate the existence and uniqueness of solutions to the system given by (36) and (37), we show that $\mathfrak{A}(U_k; U, W)$ is a coercive and continuous bilinear form and $\mathfrak{B}(W, q)$ is continuous and weakly coercive in appropriately chosen finite-element spaces.

Since $\vec{u}_k$ is a piecewise polynomial, it is bounded, as is $\nabla \vec{u}_k$. We then have the following property.

Lemma 12. *Assume that* $C_{sup} = \sup_{(x,y)\in\Omega} |\nabla \vec{u}_k|$ *, then*

$$\|\vec{v} \cdot \nabla \vec{u}_k\|_0 \leq C_{sup}\|\vec{v}\|_0, \quad \forall \vec{v} \in \mathbf{W}_h.$$

Proof: Let $N = \nabla \vec{u}_k \nabla \vec{u}_k^T$, then N is positive semidefinite, and the largest eigenvalue of N is no larger than its trace. For all vectors $\vec{y} \in \mathbb{R}^2$, we have

$$\vec{y}^T N \vec{y} \leq \lambda_{max}(N) \vec{y}^T \vec{y} \leq \mathrm{tr}(N)|\vec{y}|^2 = |\nabla u_k|^2 |\vec{y}|^2,$$

which implies that

$$\langle \vec{v} \cdot \nabla \vec{u}_k, \vec{v} \cdot \nabla \vec{u}_k \rangle_0 \leq C_{sup}^2 \langle \vec{v}, \vec{v} \rangle_0.$$

□

Theorem 10. *Let Re, Re_m be small enough such that*

$$\min\{\alpha_1 Re^{-1}, \alpha_2 Re_m^{-1}\} - (C_{sup} + D_{sup} + \frac{M_{sup}}{2}) > 0,$$

where α_1, α_2 are constants defined below, $C_{sup} = \sup_{(x,y)\in\Omega} |\nabla \vec{u}_k|, D_{sup} = \sup_{(x,y)\in\Omega} |\nabla A_k|$, and $M_{sup} = \sup_{(x,y)\in\Omega} |\vec{u}_k|$.

Then, there exists a constant $\gamma > 0$ depending on U_k and Ω such that

$$\mathfrak{A}(U_k; W, W) \geq \gamma \|W\|_1^2, \quad \forall W \in \mathbf{S}_h. \tag{38}$$

Proof: By standard arguments,

$$\langle \nabla \vec{v} + \nabla \vec{v}^T, \nabla \vec{v} \rangle_0 \geq \alpha_1 \|\vec{v}\|_1^2, \quad \forall \vec{v} \in \mathbf{W}_h,$$

where α_1 is a constant depending only on Ω (see [5], Corollary 11.2.22) and

$$\langle \nabla \psi, \nabla \psi \rangle_0 \geq \alpha_2 \|\psi\|_1^2, \quad \forall \psi \in \mathbf{X}_h,$$

where α_2 depends only on Ω (see Friedrichs' inequality [5]).

By Lemma 12,

$$|\langle (\vec{v} \cdot \nabla) \vec{u}_k, \vec{v} \rangle_0| \leq \|(\vec{v} \cdot \nabla) \vec{u}_k\|_0 \|\vec{v}\|_0 \leq C_{sup} \|\vec{v}\|_0^2, \quad \forall \vec{v} \in \mathbf{W}_h.$$

Since A_k is a piecewise polynomial function, it is bounded, as is ∇A_k. Since $D_{sup} = \sup_{(x,y)\in\Omega} |\nabla A_k|$, then,

$$|\langle \vec{v} \cdot \nabla A_k, \psi \rangle_0| \leq \|\vec{v} \cdot \nabla A_k\|_0 \|\psi\|_0 \leq D_{sup} \|\vec{v}\|_0 \|\psi\|_0.$$

Since $M_{sup} = \sup_{(x,y)\in\Omega} |\vec{u}_k|$, then,

$$|\langle \vec{u}_k \cdot \nabla \psi, \psi \rangle_0| \leq \|\vec{u}_k \cdot \nabla \psi\|_0 \|\psi\|_0 \leq M_{sup} \|\nabla \psi\|_0 \|\psi\|_0 \leq \frac{M_{sup}}{2} \|\psi\|_1^2,$$

and we have

$$\left\langle (\vec{u}_k \cdot \nabla) \vec{v}, \vec{v} \right\rangle_0 \leq \frac{M_{sup}}{2} \|\vec{v}\|_1^2, \quad \forall \vec{v} \in \mathbf{W}_h.$$

Similarly, we can bound form $\hat{a}(A_k;\vec{v},\psi)$ as,

$$|\hat{a}(A_k;\vec{v},\psi)| \leq 2D_{sup}\|\nabla\psi\|_0\|\nabla\vec{v}\|_0.$$

According to the above analysis, we have

$$\begin{aligned}
\mathfrak{A}(U_k;W,W) &\geq \alpha_1 Re^{-1}\|\vec{v}\|_1^2 + \alpha_2 Re_m^{-1}\|\psi\|_1^2 - C_{sup}\|\vec{v}\|_0^2 - D_{sup}\|\vec{v}\|_0\|\psi\|_0 \\
&\quad -\frac{M_{sup}}{2}\|\psi\|_1^2 - \frac{M_{sup}}{2}\|\vec{v}\|_1^2 - 2D_{sup}\|\nabla\psi\|_0\|\nabla\vec{v}\|_0 \\
&\geq \min\{\alpha_1 Re^{-1}, \alpha_2 Re_m^{-1}\}\|W\|_1^2 - \big(C_{sup}\|\vec{v}\|_0^2 + \frac{M_{sup}}{2}\|W\|_1^2 \\
&\quad + D_{sup}(\|\vec{v}\|_1^2 + \|\psi\|_1^2)\big) \\
&\geq \min\{\alpha_1 Re^{-1}, \alpha_2 Re_m^{-1}\}\|W\|_1^2 - (C_{sup} + D_{sup} + \frac{M_{sup}}{2})\|W\|_1^2 \\
&\geq (\gamma_1 - \gamma_2)\|W\|_1^2,
\end{aligned}$$

where $\gamma_1 = \min\{\alpha_1 Re^{-1}, \alpha_2 Re_m^{-1}\}, \gamma_2 = C_{sup} + D_{sup} + M_{sup}/2$. Let $\gamma = \gamma_1 - \gamma_2 > 0$. Thus, $\mathfrak{A}(U_k;W,W)$ is coercive. □

The following Lemmas are consequences of Lemmas 10 and 11, so we omit the proofs.

Lemma 13. *$F(U_k,p_k;W)$ and $G(U_k;q)$ are bounded linear functionals on $\mathbf{S}_h$ and $\mathbf{Q}_h$, respectively.*

Lemma 14. *$\mathfrak{A}(U_k;U,W)$ and $\mathfrak{B}(W,q)$ are continuous on $\mathbf{S}_h$ and $\mathbf{Q}_h$, respectively.*

Assumption 2. *There exists a constant $\Gamma_s > 0$ depending on Ω such that*

$$\inf_{0\neq q\in\mathbf{X}_h}\sup_{\vec{0}\neq\vec{v}\in\mathbf{W}_h}\frac{b(q,\vec{v})}{\|\vec{v}\|_1\|q\|_0} \geq \Gamma_s > 0. \tag{39}$$

Remark 1. *The major difference between (26) and (39) is that the inf-sup condition must be satisfied on the discrete space. There is, however, no restriction on the discrete space chosen to approximate A. Choosing a pair of spaces for which the discrete inf-sup condition (39) holds is well-known to be a delicate matter, and seemingly natural choices of velocity and pressure approximation do not always work [10]. For example, the simplest globally continuous approximation, using bilinear elements for both velocity and pressure (the so-called Q_1-Q_1 approximation), is unstable. In general, care must be taken to make the velocity space rich enough compared to the pressure space, otherwise the discrete solution will be "over-constrained". Any stable element pair for the Navier-Stoke equations (e.g., Q_2-Q_1 Taylor-Hood elements, Q_2-P_{-1}, Q_2-P_0) can be used for $\vec{u}$ and p (see [4, 10, 11, 12]) to satisfy (39).*

Theorem 11. *Under the assumptions of Theorem 10 and Assumption 2, there is a unique solution to Formulation 8.*

Proof: Following Theorem 1.2 of [12, Chapter III], Lemmas 10–14 prove the result. □

4.3 Solvability of stabilized discretizations

In this subsection, we give a solvability condition for stabilized finite-element methods, since our analysis is also suitable for this setting. From Formulation 8, the matrix equations that result from a stabilized finite-element discretization have the following block form:

$$\mathcal{M}x = \begin{bmatrix} F & Z & B \\ Y & D & 0 \\ B^T & 0 & -T \end{bmatrix} \begin{bmatrix} x_u \\ x_a \\ x_p \end{bmatrix} = \begin{bmatrix} f_u \\ f_a \\ f_p \end{bmatrix}, \tag{40}$$

where x_u, x_a, and x_p are the discrete Newton corrections for $\vec{u}, A$, and p, respectively, and f_u, f_a, and f_p are the corresponding blocks of the residual, while T is the stabilization term.

Let

$$\hat{F} = \begin{bmatrix} F & Z \\ Y & D \end{bmatrix}, \hat{B} = \begin{bmatrix} B \\ 0 \end{bmatrix}, x_{\hat{u}} = \begin{bmatrix} x_u \\ x_a \end{bmatrix}, f_{\hat{u}} = \begin{bmatrix} f_u \\ f_a \end{bmatrix}.$$

Then, Equation (40) can be rewritten as

$$\mathcal{M}x = \begin{bmatrix} \hat{F} & \hat{B} \\ \hat{B}^T & -T \end{bmatrix} \begin{bmatrix} x_{\hat{u}} \\ x_p \end{bmatrix} = \begin{bmatrix} f_{\hat{u}} \\ f_p \end{bmatrix}, \tag{41}$$

where $\hat{F} \in \mathbb{R}^{n\times n}, \hat{B} \in \mathbb{R}^{n\times m}, f_{\hat{u}} \in \mathbb{R}^n, f_p \in \mathbb{R}^m$ and $m \leq n$.

Lemma 15. *Under the assumptions of Theorem 2, $\hat{F}$ is positive definite.*

Proof: This is a consequence of (38). □ With homogeneous Dirichlet boundary conditions on $\vec{v} \in \mathbf{W}$, $b(p, \vec{v}) = 0$ for all $\vec{v} \in \mathbf{W}$ implies that the pressure, p, is a constant. When using a nodal finite-element basis, $\text{Span}\{\vec{1}\} \subset \text{Ker}(B)$ is a natural consequence of this. If the two spaces are equal, the resulting pressure is unique up to constants. When a discrete inf-sup condition (as in (39)) does not hold, $\text{Ker}(B) \neq \text{Span}\{\vec{1}\}$. However, we have the following condition that guarantees the solvability of the stabilized method, and gives insight into the construction of T.

Theorem 12. *Under the assumptions of Theorem 2, let $S = -(T + \hat{B}^T\hat{F}^{-1}\hat{B})$ be the Schur Complement of $\hat{F}$ in $\mathcal{M}$, with T symmetric and positive semidefinite. If $\text{Ker}(T) \cap \text{Ker}(B) \subseteq \text{Span}\{\vec{1}\}$, then $\text{Ker}(S) \subseteq \text{Span}\{\vec{1}\}$.*

Proof: Since $\hat{F}$ is positive definite, $\hat{F}^{-1}$ is also positive definite. This implies that $p^T\hat{B}^T\hat{F}^{-1}\hat{B}p \geq 0$ with equality if and only if $Bp = 0$. On the other hand, because T is symmetric positive semidefinite, $\text{Ker}(S) = \text{Ker}(T) \cap \text{Ker}(B)$. □ This theorem tells us that (41) is well-posed if the stabilized pressure Schur Complement, S, is a positive semi-definite matrix with the following stability condition:

$$\text{Ker}(S) \subseteq \text{Span}\{\vec{1}\}.$$

The important consequence of Theorem 12 is that any stabilization approach that is suitable for the Stokes equations is also suitable in this context, since

$\hat{F}$ does not enter the intersecting kernels condition. In particular, standard approaches for equal-order $Q_1 - Q_1$ approximations of velocity and pressure can be used, including diffusion stabilization and pressure-projection [9, 10]. Thus, the analysis above can be applied to discretization approaches similar to those in [8], which uses diffusion-type stabilization of the pressure equation (although we note that [8] also makes use of additional stabilization for the case when the Reynolds numbers are not small, which is not considered here). Based on the above discussions, we give the natural result.

Theorem 13. *Under the assumptions of Theorems 2 and 12, the stabilized discrete Newton approximation of Formulation 6 yields a pressure that is unique up to constants.*

4.4 Convergence of Newton's Method

Finally, we define $\|U\|_{1,\infty} := \max\{\|\vec{v}\|_{1,\infty}, \|A\|_{1,\infty}\}$ and $\mathbf{D}(U;r) = \{W : \|W - U\|_1 < r\}$. In order to give some convergence analysis of Newton's method at the discrete level, we assume the following hold

Assumption 3. *Assume the conditions of Theorem 9 hold; furthermore, assume the solution U_h^* of Formulation 6 satisfies*

$$\kappa_h^* = \|U_h^*\|_{1,\infty} < \gamma_1,$$

where $\gamma_1 = \min\{\alpha_1 Re^{-1}, \alpha_2 Re_m^{-1}\}$ is from Theorem 10.

Assumption 4. *Assume that there exists $r_1 > 0$ such that for any initial iterate $U_k \in \mathbf{D}(U_h^*; r_1)$ Newton's method converges to the unique solution of Formulation 6 and converges quadratically.*

For γ_2 from the proof of Theorem 10,

$$\gamma_2 = C_{sup} + D_{sup} + M_{sup}/2 < 3 \cdot \max\{C_{sup}, D_{sup}, M_{sup}\} < 3\|U_k\|_{1,\infty},$$

gives

$$\mathfrak{A}(U_k; W, W) > (\gamma_1 - 3\|U_k\|_{1,\infty})\|W\|_1^2.$$

Thus, if $\|U_k\|_{1,\infty} < \dfrac{\gamma_1}{3}$, then $\mathfrak{A}(U_k; W, W)$ is coercive.

Lemma 16. *Assume that $U \in \mathbf{S}_h$ and $\|U\|_{1,\infty} = \kappa_h$. Then,*

$$\|W\|_{1,\infty} \leq \kappa_h + C_1 h^{-1} r, \quad \forall W \in \mathbf{D}(U;r) \cap \mathbf{S}_h,$$

where C_1 is a constant depending on Ω.

Proof: According to the standard inverse inequality [5, Theorem IV.5.11],

$$\|U\|_{1,\infty} \leq C_1 h^{-1} \|U\|_1, \quad \forall U \in \mathbf{S}_h,$$

where C_1 is a constant. By the triangle inequality, for $W \in \mathbf{D}(U;r) \cap \mathbf{S}_h$

$$\begin{aligned} \|W\|_{1,\infty} &\leq \|U\|_{1,\infty} + \|W-U\|_{1,\infty} \\ &\leq \kappa_h + C_1 h^{-1}\|W-U\|_1 \\ &\leq \kappa_h + C_1 h^{-1} r. \end{aligned}$$

□

Remark 2. *Lemma 16 indicates that if we take $U_k \in \mathbf{D}(U;r_2)$, for $r_2 = \frac{h(\gamma_1/3-\kappa_h^*)}{C_1}$, then $\mathcal{A}(U_k;W,W)$ is always coercive.*

If for the stabilized case, we have the same approximation result as in Theorem 9, then the next convergence theorem is not only true for stable element approximations, but also for the stabilized case.

Theorem 14. *Under Assumptions of Theorem 11 or Theorem 13, and Assumptions 3 and 4, for any initial $U_0 \in \mathbf{D}(U_h^*;r^*)$, $r^* = \min\{r_1, r_2\}$, the sequence $\{U_k\}$ produced by Newton's method is both well-defined and converges to the solution of Formulation 6.*

Proof: Since $U_0 \in \mathbf{D}(U_h^*;r^*)$, then according to Lemma 16, Formulation 8 has a unique solution for every U_k. By the triangle inequality, we have

$$\|U_k - U^*\|_1 \leq \|U_k - U_h^*\|_1 + \|U_h^* - U^*\|_1. \tag{42}$$

According to Assumptions 3 and 4, (42) goes to zero. □

5 Conclusions

In this paper, we present a theoretical analysis of the weak formulations of a steady-state visco-resistive vector-potential MHD formulation. Under certain conditions, we prove the uniqueness and existence of the solutions. Furthermore, we show that the solutions of the curled and uncurled formulations are the same, under some conditions. From this point of view, using the uncurled formulation to approximate the MHD problem is reasonable and meaningful. A mixed finite-element approximation of the uncurled formulation is discussed. The convergence rates obtained under some standard smoothness assumptions have been analysed and show that it is a suitable option. Thus, using Newton stepping and a stable Stokes FEM pair plus any space for A yields a convergent solution scheme for MHD. The numerical tests in [2] demonstrate these results of the uncurled formulation.

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